

SECTION XIII

GLOBAL CONSIDERATIONS IN KIDNEY DISEASE

Global Challenges and Progress in Kidney Health

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CHAPTER OUTLINE

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KEY POINTS

- The global burden of chronic kidney disease (CKD) is rising. Impaired kidney function is the seventh leading global risk factor for death.
- In lower-income regions CKD occurs in younger individuals.
- The global burden of acute kidney injury remains unknown.
- Impaired kidney function contributes to more deaths from cardiovascular disease than from end-stage kidney failure.
- Many countries lack policies regarding kidney disease advocacy, screening, and prevention.
- Achieving universal health coverage may reduce the global burden of acute kidney injury and CKD through improved access to primary care and essential diagnostics and medication.
- Access to dialysis and transplantation is highly inequitable across the globe and within regions and countries, largely dictated by ability to pay and presence of government policies.
- If dialysis and/or transplantation are to be provided in low-resource settings, rigorous planning must be done up front to determine affordability and to ensure sustainability, and transparent criteria must be developed to achieve fairness and equitable access to kidney replacement therapy.

GLOBAL CHALLENGES IN KIDNEY HEALTH

KIDNEY DISEASE IS AN IMPORTANT PUBLIC HEALTH CONCERN

Kidney disease is an important noncommunicable disease (NCD) worldwide; however, this is not yet widely acknowledged on the global policy agenda. The World Health Organization (WHO) Global Action Plan for the Prevention and Control of NCDs initially prioritized four main diseases (heart diseases and stroke, cancer, chronic respiratory diseases, and diabetes) and four main risk factors (diet, physical activity, alcohol, and tobacco) in a 4 × 4 campaign, having identified these four diseases as the major NCD killers globally.¹ Subsequently, mental health and air pollution were added to become a 5 × 5 campaign. In 2023, oral health was included as a priority NCD, and it has been suggested that sugar be added to expand to a 6 ×

6 campaign.² Kidney disease has therefore not consistently been identified as a global priority, although it is likely that kidney health has benefited from campaigns aimed at improving adherence to healthy lifestyles.

The global importance of kidney disease as a cause of mortality and morbidity is increasing. Impaired kidney function (IKF—defined as an estimated glomerular filtration rate [eGFR] <60 mL/min/L·73 m² or albumin-to-creatinine ratio [ACR] ≥30 mg/g) has been identified as the 7th global leading cause of death (Table 73.1) and has risen to the 10th leading global cause of death as listed by the WHO.^{3,4} If this trend continues, it has been projected that kidney disease may become the fifth leading global cause of years of life lost (YLLs) by 2040.⁵

As discussed in detail later, estimates from the Global Burden of Disease (GBD) study in 2019 showed that around 1.4 million people died of chronic kidney disease (CKD) and kidney dysfunction contributed to a further 1.7 million deaths from cardiovascular diseases (CVD).³ Others have

Table 73.1 Global Burden of Chronic Kidney Disease (CKD, Mean Estimates with 95% Uncertainty Intervals)

Parameter		CKD		CVD due to IKF		IKF Total
Prevalence						
Number, 2019	697.3 mln	(650.0-741.1)	—	—	—	—
Crude rate/100,000 population, 2019	9011.9	(8401.3-9577.8)	—	—	—	—
Global rank, 2019	13th	—	—	—	—	—
Change in number from 1990 to 2019, %	104.1%	(100.9-107.3)	—	—	—	—
Change in crude rate from 1990 to 2019, %	41.1%	(38.9-43.4)	—	—	—	—
Change in age-standardized rate from 1990 to 2019, %	9.4%	(8.7-10.3)	—	—	—	—
YLDs						
Number, 2019	8.7 mln	(6.3-11.5)	2.3 mln	(1.6-3.1)	11.3 mln	(8.2-14.7)
Crude rate/100,000 population, 2019	113.0	(81.7-148.7)	29.8	(20.7-40.0)	145.8	(106.4-189.7)
Global rank, 2019	26th	—	—	—	10th	—
Change in number from 1990-2019, %	125.6%	(116.6-134.6)	119.1%	(114.1-124.4)	125.5%	(118.3-132.7)
Change in crude rate from 1990-2019, %	56.0%	(49.8-62.2)	51.5%	(48.0-55.2)	55.9%	(50.9-60.9)
Change in age-standardized rate from 1990-2019, %	24.3%	(20.6-27.7)	5.4%	(3.0-7.5)	20.2%	(16.9-23.2)
Deaths						
Number, 2019	1.4 mln	(1.3-1.5)	1.7 mln	(1.3-2.2)	3.2 mln	(2.7-3.6)
Crude rate/100,000 population, 2019	18.4	(17.0-19.7)	22.4	(17.1-28.0)	40.9	(35.2-46.8)
Global rank, 2019	11th	—	—	—	7th	—
Change in number from 1990-2019, %	137.4%	(118.7-153.6)	78.7%	(67.3-90.8)	101.2%	(88.4-112.9)
Change in crude rate from 1990-2019, %	64.1%	(51.2-75.3)	23.6%	(15.7-31.9)	39.1%	(30.3-47.2)
Change in age-standardized rate from 1990-2019, %	13.3%	(4.6-20.4)	-23.0%	(-27.5--18.5)	-10.0%	(-15.5--4.6)
YLLs						
Number, 2019	32.8 mln	(30.5-35.0)	32.4 mln	(25.3-40.1)	65.2 mln	(57.2-73.5)
Crude rate/100,000 population, 2019	423.8	(393.6-453.0)	418.9	(327.3-518.2)	842.7	(739.5-950.1)
Global rank, 2019	16th	—	—	—	10th	—
Change in number from 1990 to 2019, %	86.0%	(72.9-100.0)	66.6%	(54.7-78.2)	75.8%	(64.6-87.4)
Change in crude rate from 1990 to 2019, %	28.6%	(19.6-38.3)	15.2%	(6.9-23.2)	21.6%	(13.8-29.6)
Change in age-standardized rate from 1990 to 2019, %	2.3%	(-4.9-9.5)	-21.8%	(-27.0--16.4)	-11.3%	(-17.1--5.6)
DALYs						
Number, 2019	41.5 mln	(38.3-45.0)	34.7 mln	(27.5-42.7)	76.5 mln	(67.8-86.3)
Crude rate/100,000 population, 2019	536.9	(494.9-582.1)	448.7	(354.9-552.3)	988.5	(876.1-1115.2)
Global rank, 2019	18th	—	—	—	10th	—
Change in number from 1990 to 2019, %	93.2%	(81.6-105.0)	69.3%	(58.1-80.6)	81.7%	(71.6-92.0)
Change in crude rate from 1990 to 2019, %	33.6%	(25.6-41.7)	17.0%	(9.3-24.8)	25.6%	(18.6-32.8)
Change in age-standardized rate from 1990 to 2019, %	6.3%	(0.2-12.4)	-20.5%	(-25.5--15.5)	-7.7%	(-12.9--2.4)

IKF (impaired kidney function) includes low glomerular filtration rate and albuminuria. It is considered to be a risk factor resulting in the burden of CKD, cardiovascular disease (CVD), and gout.

Global rank for CKD refers to a rank among 369 diseases and injuries; global rank for IKF refers to a rank among 87 risk factors expressed as a crude rate.

estimated that between 2 and 7 million people died in 2010 because they had no access to dialysis and 1.7 million die annually from acute kidney injury (AKI).^{6,7} It is possible therefore that kidney disease may contribute to 6–11 million deaths annually, many of these thus far uncounted.⁸

AKI and CKD are both of public health concern. Although the true prevalence of both in low-income (LIC) and lower-middle-income countries (LMICs), collectively LLMICs, is unknown, it is estimated that they are at least as high, if not higher, than in high-income countries (HICs). Approximately 700 million adults (8%–16%) worldwide are living with CKD, around 80% of whom reside in low or middle-income countries.^{9,10} Up to 90% of those affected however are not aware of their condition.^{11,12} The prevalence of CKD among children is unknown but may reach 1%.¹³ Traditional and universal risk factors for CKD include diabetes and hypertension, although additional risk factors for CKD are being increasingly recognized including nephrotoxins (prescription medication and traditional/complementary remedies), kidney stones, fetal and maternal exposures, infections, environmental and occupational exposures, and AKI.^{14,15} The contribution of such nontraditional risk factors to the burdens of AKI and CKD is unknown but may be significant, especially in LLMIC. The impact of climate change and its consequences is also increasingly being recognized as an important contributor to both acute and chronic kidney disease.^{16–19}

Global data on the burden of AKI are lacking; however, from systematic reviews the incidence of AKI in hospitalized patients has been reported to be 7% to 21% (30%–70% of critically ill patients) worldwide.^{7,20} AKI has been estimated to affect 13.3 million people per year, of whom around 85% reside in LLMIC.²¹ Data for children are again relatively scarce, with estimates of 7.5% to 23.3% among hospitalized children depending on the definitions used.²² Overall mortality has been estimated at 11% to 23% in adults and 14% in children but varies with the population studied, reaching 80% if no dialysis is available when required.^{7,20,23} AKI is associated with high costs of care as many patients require intensive care management. Indeed, mortality is lower in countries with higher expenditure on health care, reflecting better access to health care and dialysis in these countries.^{20,24,25} In higher-income countries the majority of AKI is hospital acquired, whereas in many LLMICs, community-acquired AKI is more common.^{26,27} Sepsis, dehydration, use of nephrotoxins, and primary acute kidney injury are major universal causes of AKI, with tropical infections and obstetric causes being common in lower-resource settings.²⁷ In many patients with AKI, kidney function will recover, but adults and children experiencing AKI have an up to eightfold risk of developing CKD and are at increased risk of longer-term CVD and death.^{28–31} CKD, in turn, is an important risk factor for AKI.²⁹ For further discussion of AKI, see [Chapters 27 and 28](#).

Challenges in diagnosis of kidney disease include the necessity for access to blood and urine testing, awareness among health care workers, a degree of therapeutic nihilism in environments where access to specialist care and dialysis is limited, and the fact that kidney disease is often asymptomatic or symptoms are nonspecific until advanced stages.^{32,33} Proactive screening for early detection of AKI or CKD in high-risk individuals, to permit interventions to delay worsening of the disease, is therefore required.^{33–35} Increased awareness of risk factors for CKD and AKI in

communities and hospitalized populations is urgently needed to successfully implement prevention strategies. Kidney disease risk may begin early with low birth weight or preterm birth (see Chapter 20) and accumulate across the life span, often exacerbated by the social determinants of health.^{36,37} Kidney disease also bridges communicable and noncommunicable diseases. Acute infections such as malaria and leptospirosis may lead to AKI, and chronic infections such as hepatitis B and C and HIV are important causes of CKD.^{14,38} In turn, AKI and CKD increase the risk and morbidity of infections.³⁹ Many people worldwide are therefore affected by kidney disease, at least some of which is preventable.^{40,41} Despite the best efforts at prevention, however, some people will develop either acute or end-stage kidney failure and require kidney replacement therapy (dialysis or transplantation, KRT) to remain alive.

The public health importance of CKD lies in not only the risk of end-stage kidney failure (ESKF) requiring dialysis and/or transplantation, at great expense to the individual and the health system, but also in the fact that CKD is a major risk multiplier of cardiovascular risk, contributing to excess strokes and heart attacks and leading to death of many CKD patients before they ever reach ESKF.⁴² It is frequently underappreciated that CKD alone is a stronger risk factor for coronary events than diabetes, and when the two conditions coexist (which occurs in one in three patients with diabetes), the risk of cardiovascular events and overall mortality is further multiplied.^{43,44} In addition, people living with kidney disease are among those with the greatest multimorbidity and have a high degree of complexity, requiring frequent hospitalizations, and at significant cost almost equaling that of KRT.^{35,45}

In recent years various initiatives have been launched or supported by the global nephrology community to improve understanding of the burden of kidney disease, estimate the resources available to tackle kidney disease, advocate for kidney health, and contribute to the development of innovative strategies to prevent or detect and treat kidney disease early. Some of these initiatives are described as follows.

GLOBAL BURDEN OF DISEASE STUDY—KIDNEY DISEASE

The Global Burden of Disease, Injuries and Risk Factors (GBD) study presents a comprehensive assessment of the global epidemiology of CKD as both a direct cause of mortality and morbidity and a risk factor for CVD. The CKD metrics are available at the global and regional levels, as well as for 204 countries and territories from 1990 to 2019.⁴⁶ Compared with previous estimates, the latest GBD revision covering 2019 provides data for all CKD stages burden (previously limited to stages 3–5) and evaluates both albuminuria and low GFR as risk factors under the umbrella term “impaired kidney function (IKF)” (previously limited only to low GFR). The GBD uses the term “IKF” to indicate the impact of CKD as a risk factor for ischemic heart disease, stroke, peripheral vascular disease, and gout, as well as to differentiate it from the term “CKD” itself, which refers to a direct cause of morbidity and mortality due to kidney disease.¹⁰

For CKD and other diseases, the GBD provides several innovative metrics including years lived with disability (YLDs), YLLs due to premature death, and disability-adjusted life years (DALYs). DALYs are a unified measure

of disease burden that summarizes YLLs and YLDs, thus accounting for both mortality and morbidity. For each metric both crude (all ages) and age-standardized rates per 100,000 population are available. Age standardization was done to the 2019 world population, reporting rates adjusted to differences in age structure, and thus allows comparison of rates in different countries and across years for the same country. Generally, crude rates are higher in HICs, mostly due to the longer life duration and higher proportion of older individuals, while age-standardized rates are much higher in LMICs, indicating more young individuals of working age affected by CKD.

CHRONIC KIDNEY DISEASE PREVALENCE

Global CKD prevalence amounted to 697.3 million in 2019 (see Table 73.1, Fig. 73.1) which corresponds to 9.0% of the world's population. This estimate is higher than the prevalence of CVD, neoplasms, diabetes mellitus, chronic obstructive pulmonary disease, osteoarthritis, or the number of people living with the consequences of road injury (Fig. 73.2).^{10,47} CKD prevalence was 13.1% of global population aged 20 years and older and substantially increased with age. CKD was more prevalent in females (382.7 million) than in males (314.3 million).¹⁰ The majority of CKD cases in 2019 were assumed to be due to intrinsic or unknown etiology of the kidney disease (531.1 million), followed by diabetic nephropathy (134.6 million) and hypertensive nephropathy (31.0 million); however, these estimates are based on data from a minority of world countries and require data on CKD etiology from LMICs to become more precise.⁴⁸

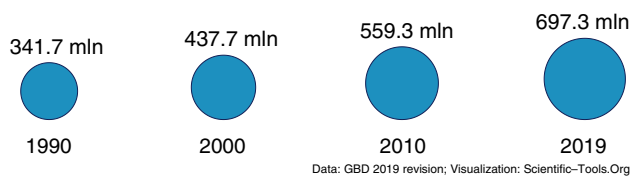


Fig. 73.1 Global chronic kidney disease prevalence, 1990–2019. Numbers on the top indicate global chronic kidney disease prevalence. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

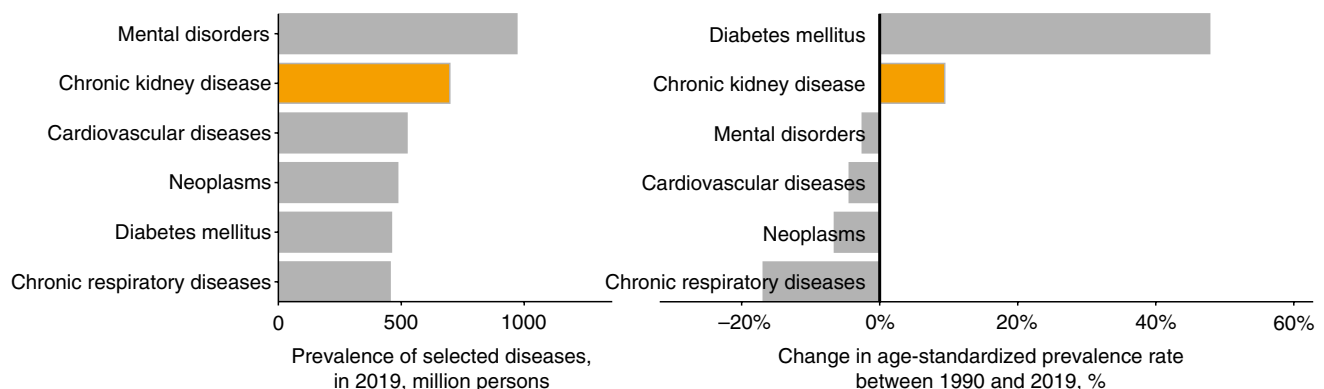


Fig. 73.2 Comparison of global prevalence for selected noncommunicable diseases. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

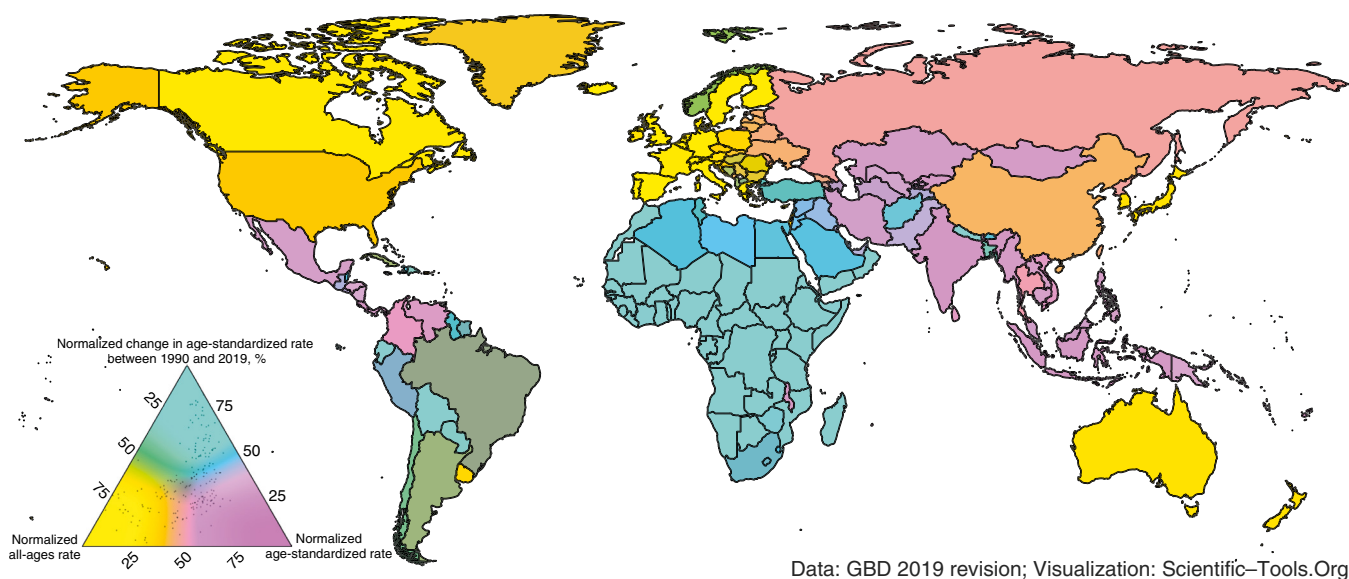


Fig. 73.3 Country-level chronic kidney disease prevalence metrics. Color filling accounts for three parameters: all-ages and age-standardized rates in 2019 and age-standardized rate change since 1990. Before mapping, each parameter was normalized to the scale from 0 to 100 (representing lowest and highest value of the original parameter). This allows visualization of all parameters and identification of the predominant one for each country. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

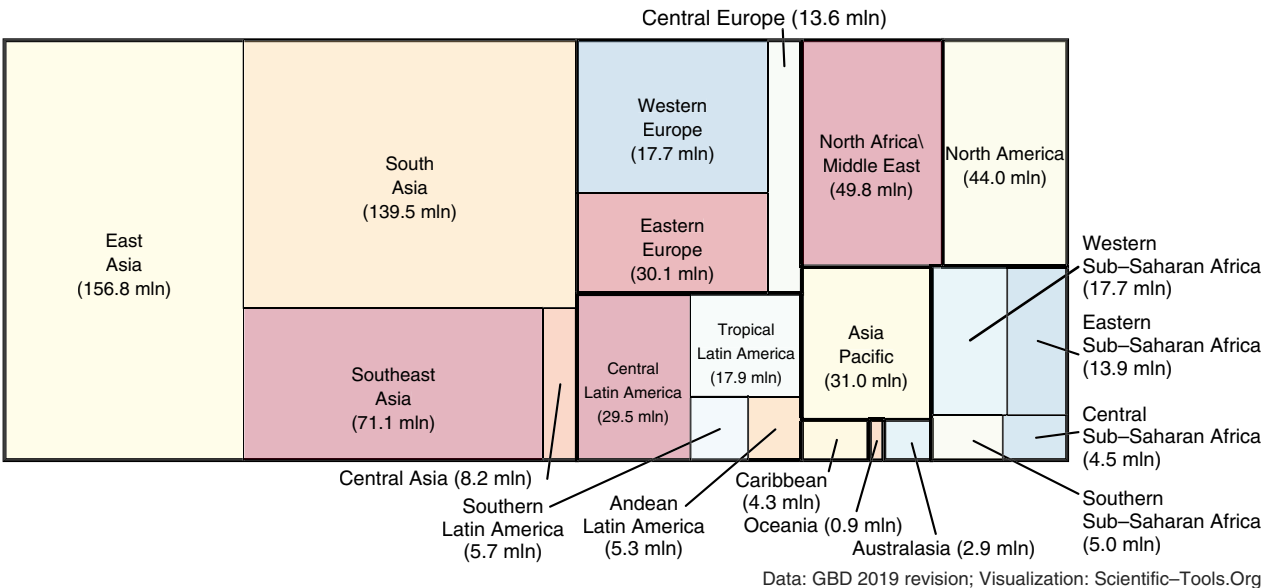


Fig. 73.4 The absolute number of chronic kidney disease patients in the world regions in 2019. Area size is proportional to number of patients in the region that also presented in the brackets. Color filling reflects age-standardized rates. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

access to and quality of kidney care exist between different countries,^{50,51} and even within the same country different populations have higher risks of late referral to nephrology services.⁵²⁻⁵⁴ Taken together, the higher frequency among younger individuals, limited access to kidney care, and inadequate access to essential medications contribute to the CKD burden, which has important implications for national economies, especially in countries with lower socioeconomic development.^{41,55,56}

MORTALITY DUE TO CHRONIC KIDNEY DISEASE

Importantly, kidney disease represents both a direct cause of morbidity and mortality and is a risk factor for morbidity and mortality from other diseases.⁵⁷ Worldwide in 2019, CKD directly resulted in 1.4 million deaths, and 1.7 million deaths occurred due to IKF as a risk factor for CVD (Fig. 73.6). Thus the overall mortality from IKF was 3.2 million, which makes it the seventh leading cause of death that resulted in

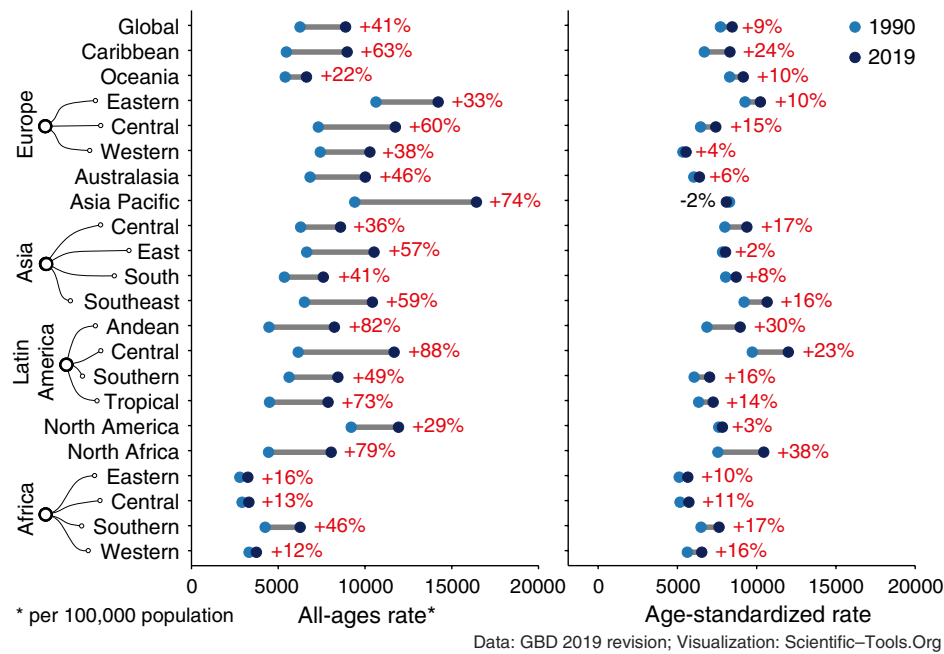


Fig. 73.5 Chronic kidney disease all-ages and age-standardized prevalence rates. Age standardization accounts for the differences in the regional age structure and used the world 2019 GBD population. “North Africa” category reflects data for North Africa and Middle East. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

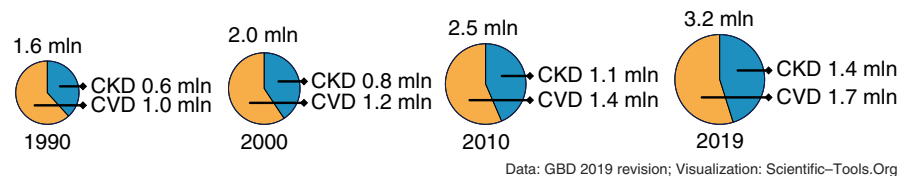


Fig. 73.6 Global mortality related to impaired kidney function, 1990–2019. Cardiovascular disease (CVD) mortality numbers reflect only a fraction of CVD mortality related to impaired kidney function. Numbers on the top indicate total number of deaths from both CKD and CVD due to impaired kidney function. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

1 in 17 fatal outcomes in the world (see Table 73.1). Of note, compared with 1990, the dynamics of the two IKF mortality components were different. The first component, mortality from CKD as a direct cause of death, increased by 137.4% in absolute numbers, by 64.1% in crude rates, and by 13.3% in age-standardized rates (see Table 73.1).⁵⁷ The highest rates and greatest increase in CKD mortality over 30 years were detected in Latin America and Southern Africa, while in some regions the improved access to KRT for older people led to a decrease in age-standardized rates (Figs. 73.7–73.9). While the number of deaths from CKD is much lower than from CVD and comparable with those for diabetes, CKD remains one of the few causes of death that is still increasing (Fig. 73.10). This increase made CKD the 11th leading cause of death in 2019 according to the GBD study, compared with its ranking of 19th in 1990, and it is expected to increase further by 2040 to 2.2 million in a best-case scenario and up to 4.0 million in a worst-case scenario, to become the fifth leading cause of YLLs.⁵ In contrast, the second component, CVD deaths attributable to IKF as a risk factor, decreased by –23.0% in age-standardized rates because of widespread implementation of prevention programs aimed to fight CVD.

A different approach to estimation of CKD mortality, based on the comparison of actual provision of KRT and

estimated need for KRT suggested that between 2.3 and 7.1 million deaths worldwide could be attributable to death from ESKF due to lack of access to KRT.⁶ Determining the exact number of deaths related to CKD is challenging because of the lack of access to diagnosis of CKD in many lower-income regions, differences in mortality coding practices, competitive categorical mortality attribution (CKD vs. CVD), and lack of precise data from many countries. Nevertheless, even with the available data, the mortality burden from kidney disease is huge and requires urgent implementation of CKD prevention and treatment programs everywhere.

YLDS, YLLS, DALYS DUE TO CHRONIC KIDNEY DISEASE

Worldwide in 2019, YLDs directly related to CKD accounted for almost 8.7 million, and those related to CVD and gout attributable to IKF accounted for 2.3 million and 0.2 million, respectively (see Table 73.1). This corresponds to 1 of every 76 global YLDs from all diseases, an important number, considering that the vast majority of individuals with CKD stages 1 to 3 are not included in the YLDs due to the GBD methodology.⁴⁷ Regarding the burden expressed in YLLs, the impact reached 32.8 million for CKD as a

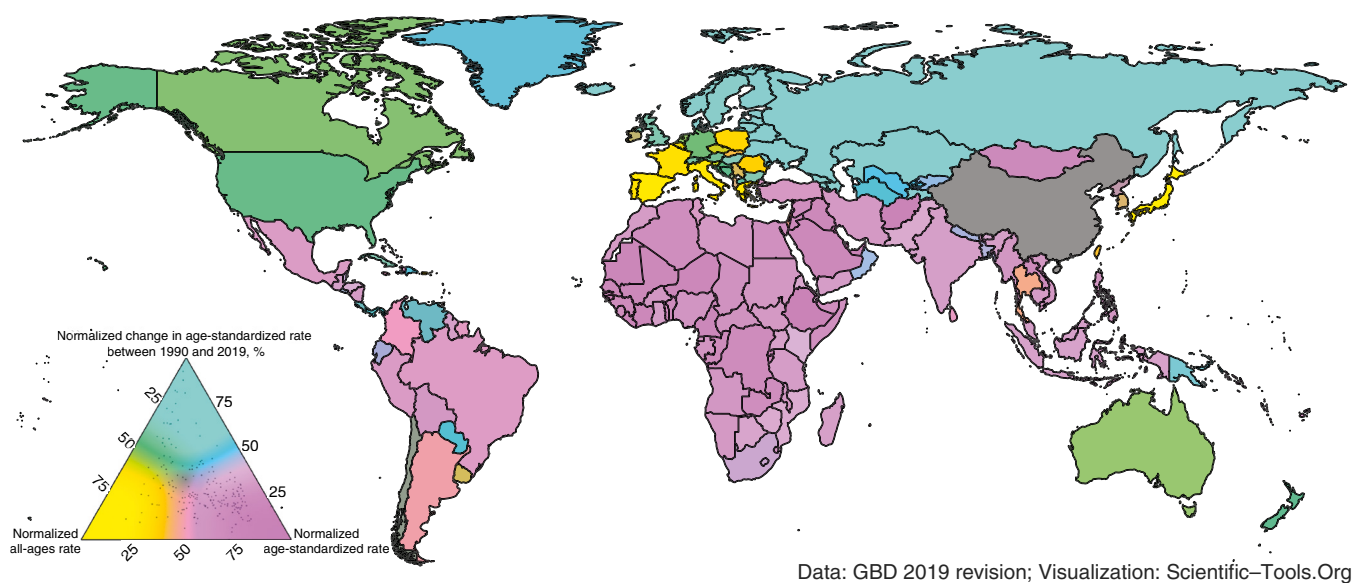


Fig. 73.7 Country-level mortality metrics for chronic kidney disease (CKD) as the direct cause of death. CKD metrics consider only the direct cause of death and do not include CVD mortality due to impaired kidney function. Color filling accounts for three parameters: all-ages and age-standardized rates in 2019 and age-standardized rate change since 1990. Before mapping each parameter was normalized to the scale from 0 to 100 (representing lowest and highest value of the original parameter). This allows to visualize all parameters and to identify the predominant one for each country. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

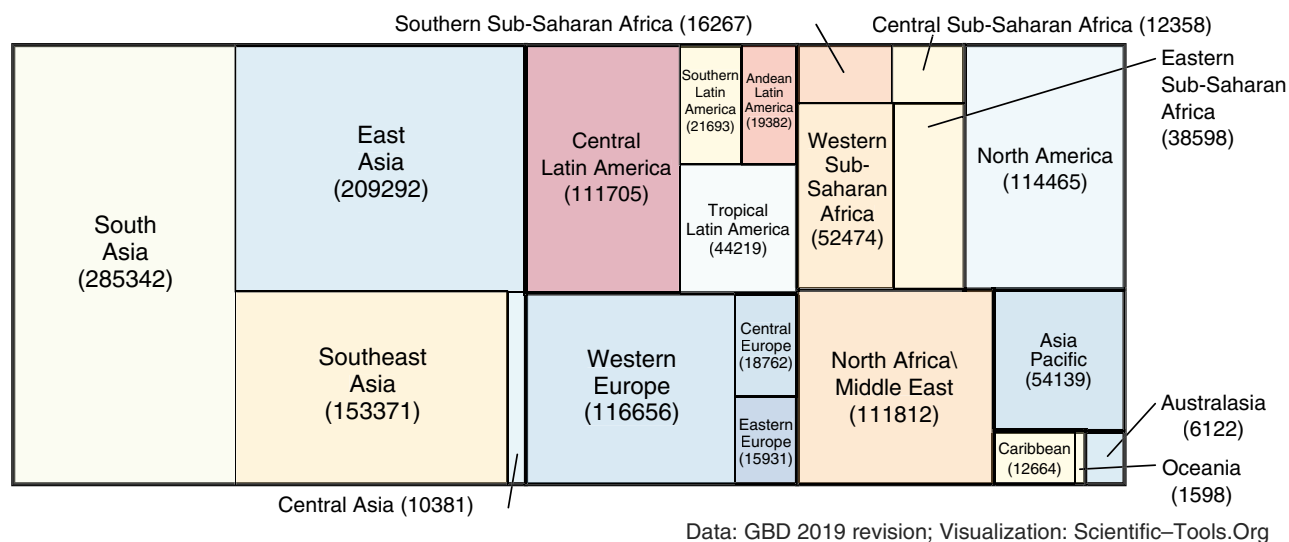
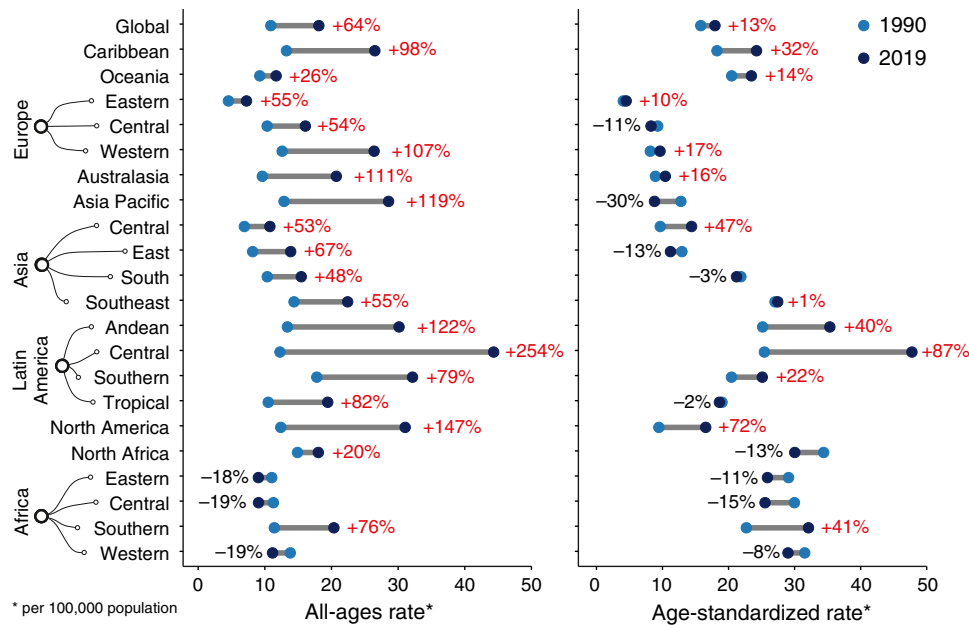


Fig. 73.8 The absolute number of deaths due to chronic kidney disease (CKD) in 2019. Area size is proportional to number of deaths in the region that is also presented in the brackets. Color filling reflects age-standardized rates. The numbers and rates do not account for impaired kidney function as a risk factor for cardiovascular mortality and reflect only CKD as a direct cause of death. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

direct cause of death and 32.4 million for CVD deaths attributed to IKF as a risk factor. Together, this equals 65.2 million YLLs, which makes IKF the 10th leading cause of YLL globally.⁵⁷ Both YLDs and YLLs substantially increase with age (Fig. 73.11).

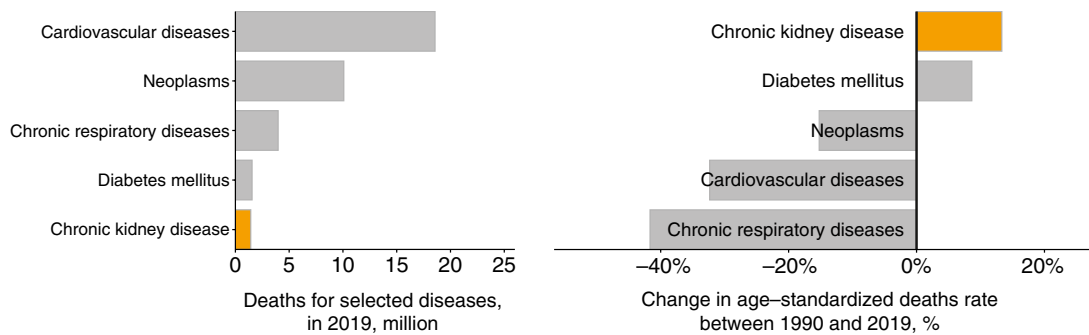
In terms of DALYs, CKD directly led to 41.5 million and IKF as a risk factor for CVD and gout led to 34.7 million and 0.2 million, respectively (Fig. 73.12).

Overall, this indicates that 1 of every 33 DALYs in the world was due to IKF, making it the 10th leading risk factor for DALYs.⁵⁷ Considering both absolute count, crude, and age-standardized DALY rates per 100,000 population, IKF outranked factors such as alcohol use, unsafe sanitation, low physical activity, drug use, and many nutritional deficiencies.⁵⁷ Considering the change in age-standardized DALY rates between 1990 and 2019,



Data: GBD 2019 revision; Visualization: Scientific-Tools.Org

Fig. 73.9 Chronic kidney disease (CKD) all-ages and age-standardized mortality rates. Age standardization accounts for the differences in the regional age structure and was made using the world 2019 GBD population. CKD metrics consider only the direct cause of death and do not include CVD mortality due to impaired kidney function. “North Africa” category reflects data for North Africa and Middle East. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)



Data: GBD 2019 revision; Visualization: Scientific-Tools.Org

Fig. 73.10 Comparison of global mortality from selected noncommunicable diseases. Reflects only the direct mortality due to disease and does not account for the impact of risk factors. Chronic kidney disease metrics consider only the direct cause of death and do not include CVD mortality due to impaired kidney function. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

there was an increase of 6.3% for CKD and a decrease of -20.5% for CVD due to IKF. The regional and country dynamics were heterogeneous, and the increase in age-standardized DALY rates over 30 years were highest in the regions with the highest CKD burdens—Central and Andean Latin America, the Caribbean, Southern Africa, and Oceania. These regions require focused efforts to reduce the excessive CKD burdens.

The GBD provides some data on the risk factors for the development of CKD (Fig. 73.13). Although these estimates are not complete due to lack of data from many countries, they do help to understand the global landscape and prioritize targeting of the major risk factors, such as high systolic blood pressure, high fasting plasma glucose, and high body mass index.⁵⁷

DRIVERS OF EPIDEMIOLOGIC TRANSITION FOR CHRONIC KIDNEY DISEASE

The change in CKD epidemiology in recent decades has occurred because of various factors. Firstly, CKD morbidity and mortality are related to the global processes of population growth, aging, and migration. The world's population has grown from 5.3 billion in 1990 to 7.7 billion in 2019, median global age increased from 23 to 30 years,⁵⁸ life expectancy increased from 65 to 73 years, and the percentage of urban population changed from 43.0% to 55.2% in the same period.⁵⁹ These changes directly impact the epidemiology of CKD, with absolute counts of CKD patients increasing in proportion to population growth and aging, and possibly also to improved health outcomes in the urban populations.

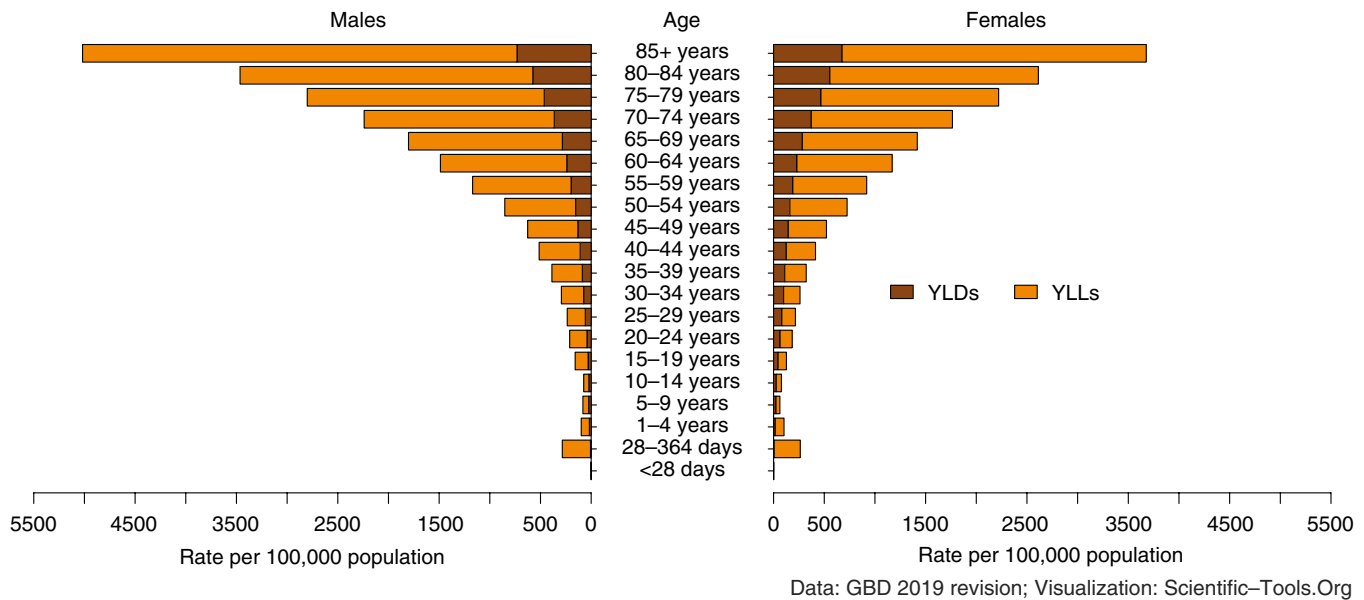


Fig. 73.11 Years of life lost, years lived with disability, and disability-adjusted life-year (DALY) rates/100,000 population for chronic kidney disease (CKD) in 2019. Orange bars represent YLLs; blue bars represent YLDs; total size of the bars equal to DALYs. (Adapted from Bikbov B. Core global metrics of years lived with disability [YLDs], years of life lost [YLLs], and disability-adjusted life years [DALYs] due to impaired kidney function [IKF] in the GBD 2019 Study. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

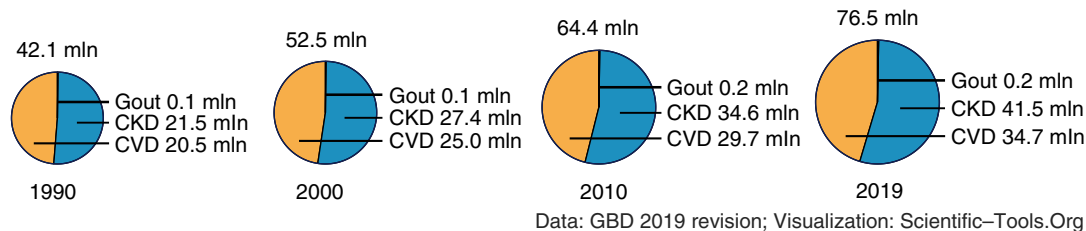


Fig. 73.12 The absolute number of disability-adjusted life years (DALYs) related to impaired kidney function. Cardiovascular disease (CVD) mortality numbers reflect only a fraction of CVD mortality related to impaired kidney function. Numbers on top indicate total number of DALYs from both chronic kidney disease (CKD), CVD, and gout due to impaired kidney function. (Adapted from Bikbov B. Core global metrics of years lived with disability [YLDs], years of life lost [YLLs], and disability-adjusted life years [DALYs] due to impaired kidney function [IKF] in the GBD 2019 Study. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

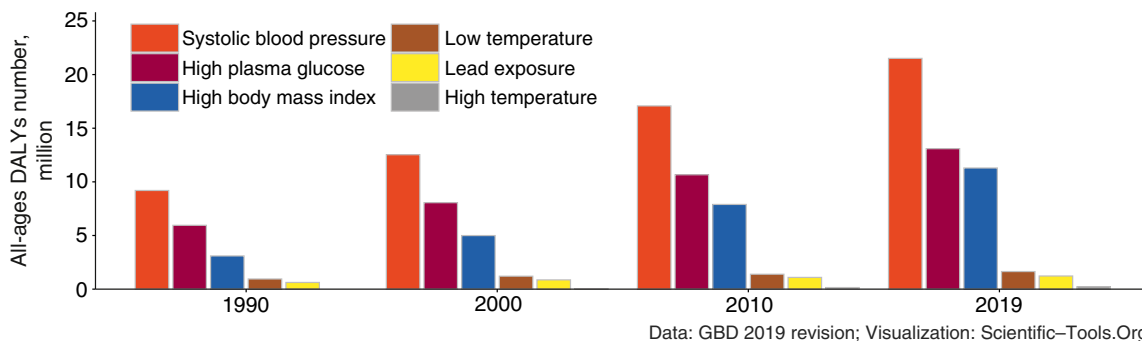


Fig. 73.13 Disability-adjusted life years for risk factors responsible for chronic kidney disease development. (Adapted from Bikbov B. Core global metrics of years lived with disability [YLDs], years of life lost [YLLs], and disability-adjusted life years [DALYs] due to impaired kidney function [IKF] in the GBD 2019 Study. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

This epidemiologic transition also drives the metabolic and nutritional risk factors for CKD development and progression. Thus over 29 years, the CKD age-standardized DALY rates attributable to arterial hypertension have increased by 15.0%, to high fasting plasma glucose by 12.6%, to high body mass index by 78.1%.⁵⁷ During the same period,

however, the DALY rates attributable to these risk factors have decreased in the case of CVD.

A second factor impacting CKD epidemiology is the absence of universal health coverage (UHC) in many LMICs or lack of access to KRT even in countries formally considered to have UHC. The lack of access to essential

medicines for almost 2 billion persons in the world leads to inability to correct risk factors and treat early CKD stages, which fuels morbidity.⁶⁰ A shortage of nephrology services results in late referral and higher rates of complications and mortality.³⁷ The lack of universal access to life-saving KRT fuels mortality, and the shortage of kidney transplant facilities increases the total cost of maintenance of KRT. The world's most socially vulnerable regions suffer the most because even after the start of dialysis, a vast majority of patients are forced to withdraw from this life-saving treatment (up to 85% of the incident patients) and die because of inability to pay for KRT.^{6,61}

A third important driver of CKD growth, especially relevant to the increase in LMICs, is the persistent inequality among different population subgroups in exposure to risk factors, as well as access to prevention and treatment.³⁷ Countries with lower economic development have higher rates of CKD and NCDs in general.¹⁰ However, even within the same country substantial inequalities exist in access to medical care, particularly KRT.^{62,63} Thus in many regions medical care for rural populations is insufficiently developed, and one would expect a higher mortality from CKD and ESKF even if the prevalence of CKD risk factors might be lower in rural settings. CKD is also more prevalent among the urban poor than those of higher socioeconomic status in developed countries⁶⁴ (see Chapter 81). Other vulnerable groups include women, older persons, and children who in many societies still have reduced access to medical treatment. Women have higher CKD prevalence and generally lower rates of kidney disease progression as compared with men, but there is also evidence of lower access to kidney treatment for women.^{65,66} Moreover, CKD in pregnant women substantially increases the odds of preeclampsia, preterm delivery, small for gestational age, and low birth weight, all factors associated with low nephron number and increased risks of CKD and other non-communicable diseases over the life course (see Chapter 20).^{67,68} Thus persisting inequalities and lack of attention to women's health forms a vicious circle that contributes to the CKD burden over generations. It could be speculated that current higher age-standardized rates of CKD burden in LLMIC may in part be impacted by suboptimal fetal and early childhood nutrition and development leading to later-life CKD.⁶⁹ Future generations may see the impact of the current humanitarian crises scattered over all continents on fetal health and CKD epidemiology over the coming decades.⁷⁰

A fourth reason underlying the CKD epidemiologic transition is the increase in CKD burden in certain countries and whole regions independent of changes in the general population. Significant increase in age-standardized CKD rates across regions highlight the most problematic countries where investigation is urgently needed to examine and address the factors leading to increasing age-standardized rates. An example of such an emerging CKD epidemic is Mesoamerican nephropathy (more probably a group of nephropathies also termed CKD of unknown etiology, CKDu), which affects predominantly adult agricultural workers in subequatorial countries of Central America and South Asia. The available data have not permitted an estimate of the burden of CKDu over time, but in some regions such as Nicaragua, CKD is the number 1 cause of death, fueled by the epidemic of CKDu.⁷¹

See Chapter 83 for a more detailed discussion of CKDu. Additional evidence, not well represented in the scientific literature, suggests that other groups of workers (metal and diamond miners, industrial paintshop workers, and others) with high exposure to environmental or occupational hazards are also affected by nephropathies that require further investigation and efforts at prevention.⁷²⁻⁷⁷ It is possible that worldwide, millions of people work or live in unsafe environments, which add to the global CKD burden.

Finally, CKD prevalence estimates depend on the precision of eGFR equations (i.e., the correspondence between measured and estimated GFR) and availability of repetitive testing to confirm chronic persistence of abnormal kidney disease markers. Use of different GFR estimation equations may lead to an up to 30% difference in CKD prevalence estimates.^{78,79} The GBD accounts for this and applied an adjustment for different eGFR equations by a special cross-walk procedure, setting the CKD-EPI (version 2009) as the reference.⁷⁸ However, any eGFR equation inherits a substantial imprecision and requires verification in different ethnic and age subgroups, which so far has not been performed in many nations suffering from the highest CKD rates, and should be implemented in the future.

THE INTERNATIONAL SOCIETY OF NEPHROLOGY GLOBAL KIDNEY HEALTH ATLAS

As part of the “Closing the Gaps” initiative of the International Society of Nephrology (ISN), the Global Kidney Health Atlas (ISN-GKHA) has provided an overview of the current state of kidney care and how it is organized and structured across the world, focusing on existing kidney care (AKI, CKD) awareness levels (captured based on existence of relevant policies and guidelines), care structures, and workforce for optimal care delivery.^{51,80,81} Three iterations of the project have so far been completed in the years 2017, 2019, and 2023. At inception, a total of 125 United Nations Member States (encompassing 93% of the world's population) responded to the survey as described in more detail elsewhere.⁸² For comprehensiveness, the elements of the survey were evaluated within the framework of the Health Systems building blocks outlined by the WHO: leadership and governance, health care financing, health workforce, medical products and technologies, information and research, and service delivery. Study respondents across countries included national nephrology society leadership, policy makers, and patient organization representatives, identified by the country and regional nephrology leadership through the ISN.⁸²

INFRASTRUCTURE FOR TREATMENT OF ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

The ISN-GKHA is the first initiative to assess global capacity for kidney care in terms of the key building blocks of a functional health system and to evaluate the readiness of countries and regions to enhance such care. There were significant gaps in services, workforce, and capacity for

research in some countries and regions. For example, most countries in Africa had no facilities for peritoneal dialysis (PD) or kidney transplantation (KT). Few countries had public funding for KRT services and medications for CKD care (including dialysis and transplant); there was a large private (out-of-pocket) contribution toward payment for KRT services and medications, particularly in countries across Africa, South Asia, Oceania, and Southeast Asia. More than two-thirds of countries reported the absence of capacity to participate in clinical research. The noted deficits in infrastructures for kidney care have shown variability with each iteration. The first iteration reported in 2017 showed that globally, 95%, 76%, and 75% of countries had facilities for hemodialysis (HD), peritoneal dialysis (PD), and KT.⁸² Although in 2019, the second iteration showed 100%, 76%, and 74% of countries had HD, PD, and KT, respectively, LICs and LMICs had much lower availability of these important services.⁸³ The third iteration, reported in 2023, using a similar survey instrument showed massive gaps in availability and access to all forms of KRT services between high-income and lower-income countries.⁵¹

AVAILABILITY OF CHRONIC KIDNEY DISEASE MONITORING AND MANAGEMENT SERVICES

There was wide variation in terms of availability of services for CKD monitoring and management at the primary and secondary/tertiary care levels across countries based on income groups (Fig. 73.14). LMIC, especially in Africa, had limited services for the diagnosis, management, and monitoring of CKD at the primary care level, with only 12% having serum creatinine measurement including eGFR. Twenty-nine percent of LIC had access to qualitative

urinalysis using urine test strips; however, no LIC had access to urine albumin-to-creatinine ratio (UACR) or urine protein-to-creatinine ratio (UPCR) measurements at the primary care level. Across all income groups, availability of services at the secondary/tertiary care level was considerably higher compared with the primary care level.

AVAILABILITY OF KIDNEY REPLACEMENT THERAPY SERVICES

Dialysis

All countries reported having chronic HD services, and more than 90% of countries reported having acute HD. More than 90% of upper-middle and HICs reported having chronic PD services, with 64% and 35% availability in LMIC and LIC, respectively. In comparison, acute PD had the lowest availability across all income groups.

KIDNEY TRANSPLANTATION

More than 90% of upper-middle and HICs reported having kidney transplant services, with more than 85% of these countries reporting the organ source as a combination of living and/or deceased donors. As expected, LIC had the lowest availability of kidney transplant services, with only 12% reporting availability and live donors as the only source.

WORKFORCE FOR CARE DELIVERY

Considerable variation was noted in the density of nephrologists across countries (Fig. 73.15). The lowest density (<5 nephrologists per million population) was common in LIC, whereas the highest density (>15 nephrologists per million population) was mainly reported in HIC. Most countries

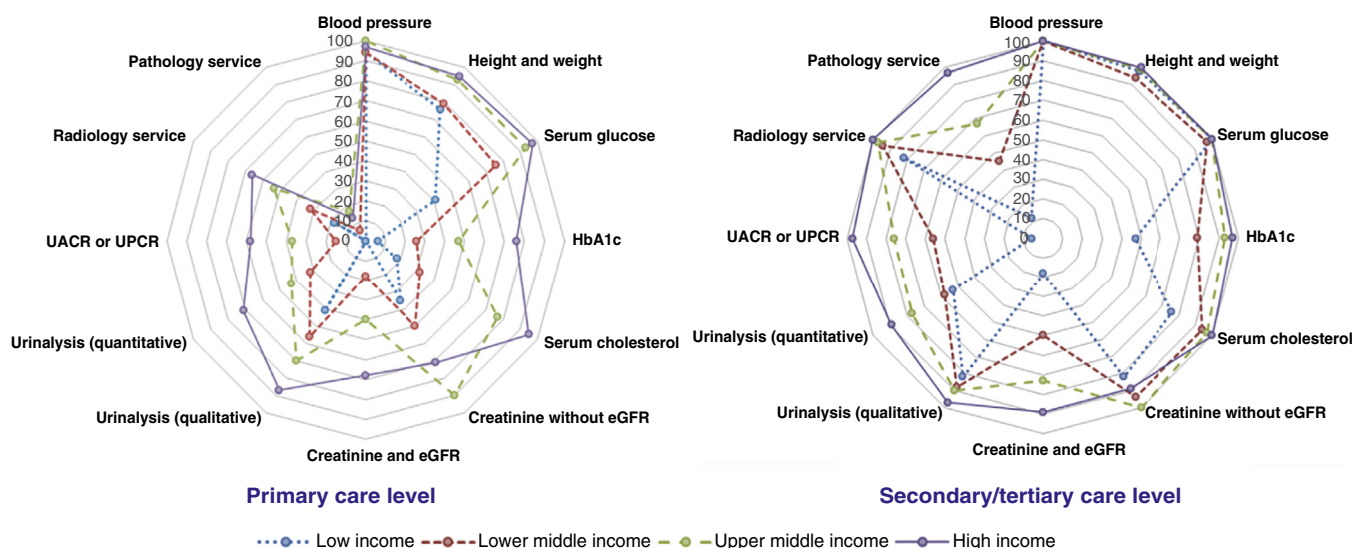


Fig. 73.14 Health care services for the identification and management of chronic kidney disease at primary care and secondary/tertiary care levels by World Bank income groups. Capacities of health care services for chronic kidney disease care are reported as percentages of countries with particular services in each income group. eGFR, Estimated glomerular filtration rate; HbA1c, glycated hemoglobin; UACR, urine albumin-to-creatinine ratio; UPCR, urine protein-to-creatinine ratio. (Adapted with permission from Htay H, Alrukhami M, Ashuntantang GE, et al. Global access of patients with kidney disease to health technologies and medications: findings from the Global Kidney Health Atlas project. *Kidney Int Suppl.* 2011;8[2]:64–73, 2018.)

reported nephrologists as primarily responsible for both CKD and AKI care. Primary care physicians (PCPs) had more responsibility for CKD care than AKI care, as 64% of countries reported PCPs primarily responsible for CKD and 35% for AKI. Intensive care specialists were primarily responsible for AKI in 75% of countries, understandable as AKI is an acute condition typically treated in hospital. However, only 45% of LICs reported that intensive care specialists were primarily responsible for AKI, compared with 90% of HICs. This discrepancy may be due to a general shortage of intensive care specialists in LIC. The appropriate number of nephrologists in a country depends on many factors including need, priority, and resources, and as such there is no global standard with respect to nephrologist density. Regardless, the density in LIC suggests a shortage of nephrologists, which is problematic as nephrologists are essential to provide leadership and a lack of nephrologists may result in consequences for policy and practice. Encouragingly, the number of nephrologists and nephropathologists is rising in LIC, in part due to the existence of fellowship programs supported by international nephrology organizations.^{84,85} Importantly, the role of a nephrologist may differ depending on how the health care system is structured. In some regions, kidney disease patients are primarily cared for by PCPs, and as such the need for nephrologists may differ compared with other regions that primarily depend on a nephrologist for managing kidney disease patients.

The density statistic merely represents the number of nephrologists per million population and provides

no indication of the adequacy to meet the needs of the population or quality of care, which depends on volume of patients with kidney disease and workforce support. For the other care providers essential for kidney care, there were variations in distribution (availability and adequacy) across countries. Overall, provider shortages were highest for renal pathologists (86% of countries reported a shortage), vascular access coordinators (81%), and dietitians (78%) and were more common in low-income countries. Fewer countries (35%) reported a shortage of laboratory technicians (Fig. 73.16).

PROGRESS IN THE ORGANIZATION, STRUCTURES, AND PROCESSES OF CARE AT THE GLOBAL LEVEL, PREPANDEMIC AND POST PANDEMIC

CHANGES IN THE AVAILABILITY OF HEALTH TECHNOLOGIES FOR THE CARE OF PATIENTS WITH KIDNEY FAILURE

Globally, the number of centers for the provision of HD is reported to have increased by 9.8%, rising from 4.4 per million population (pmp) in 2019 to 4.8 pmp in 2023 (Fig. 73.17). This proportion increased across most regions except the Middle East, where there was a decrease of -13.1% (3.8 pmp in 2019 to 3.3 pmp in 2023). The number of centers providing PD across countries increased overall by 13% (1.4–1.6 pmp). However, there was a noted decrease in the number of PD centers in Eastern

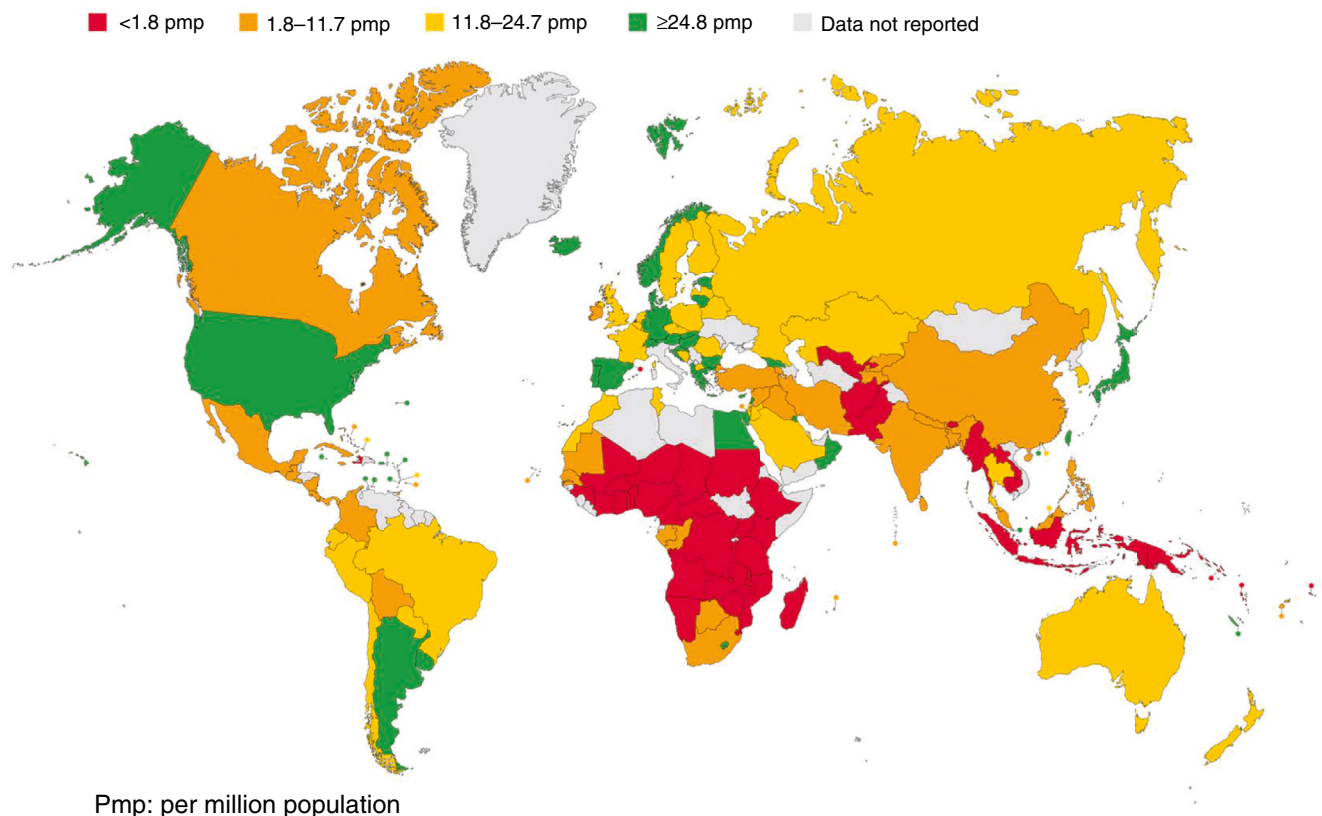


Fig. 73.15 Distribution of the global nephrologist workforce. The map depicts global distribution of nephrologists per 1 million population by country and region. Data not available indicates that data were either not known or not provided on the questionnaire for countries that received the survey. (Adapted with permission from Bello AK, Levin A, Tonelli M, et al. Assessment of global kidney health care status. *JAMA*. 2017;317[18]:1864–1881.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

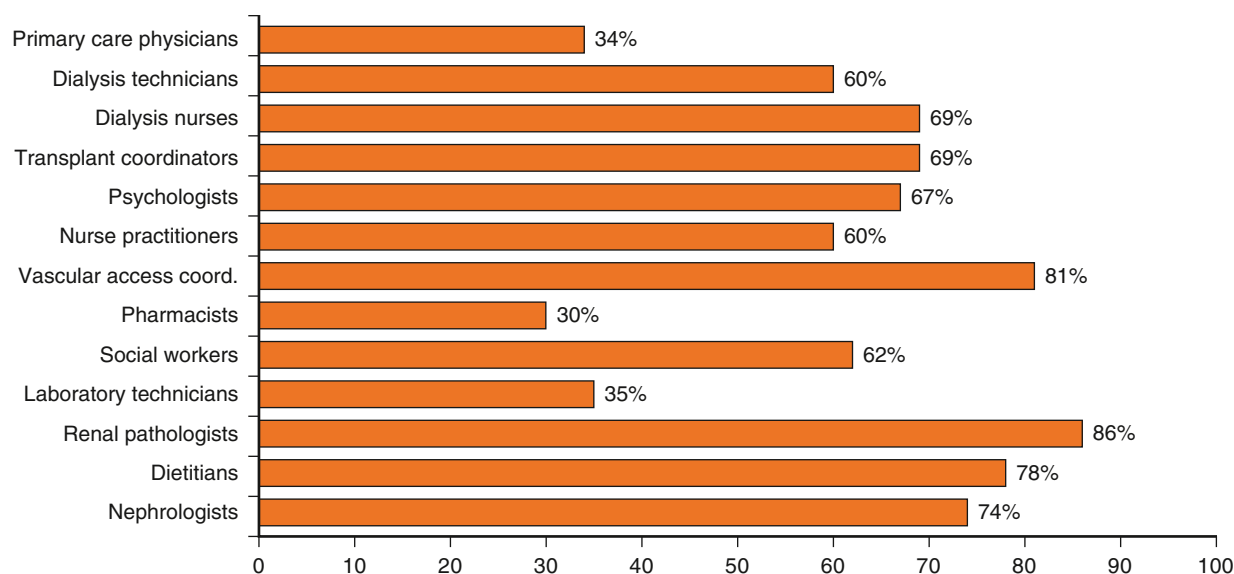


Fig. 73.16 Overall shortage in nephrology workforce. (Data from Global Kidney Health Atlas https://www.theisn.org/images/ISN_Biennial_Report_2011-2013/GKHAtlas_Linked_Compressed1.pdf)

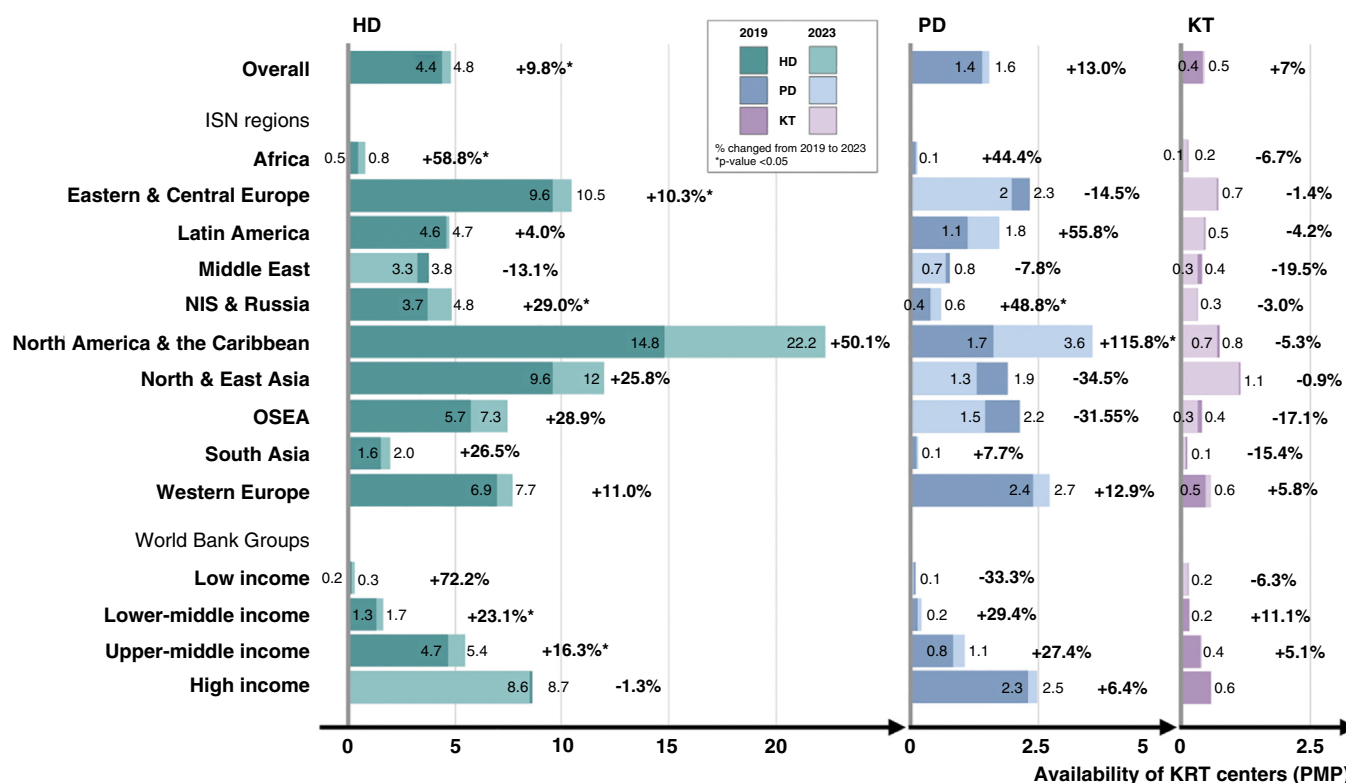


Fig. 73.17 Availability of kidney replacement therapy (KRT) centers by International Society of Nephrology (ISN) regions and World Bank Groups, pre (2019) and post pandemic (2023). (Prevalence of hemodialysis (HD), peritoneal dialysis (PD), and kidney transplantation (KT) centers in 2019 and 2023. Trends presented in percent change from 2019. *NIS*, Newly independent states; *OSEA*, Oceania and South East Asia; *PMP*, per million population.)

and Central Europe (−14.5%), the Middle East (−7.8%), North and East Asia (−34.5%), and Oceania and South East Asia (OSEA) (−31.5%) (see Fig. 73.17). Globally, the number of centers providing KT was reported to have increased by 7% (0.43–0.46 pmp). Region-specific trends

showed these centers only increased in Western Europe from 0.52 pmp in 2019 to 0.55 pmp in 2023, representing an increase of 5.8%, but evidenced a reduction in other regions, ranging from −0.9% in North and East Asia to −19.5% in the Middle East region (see Fig. 73.17).

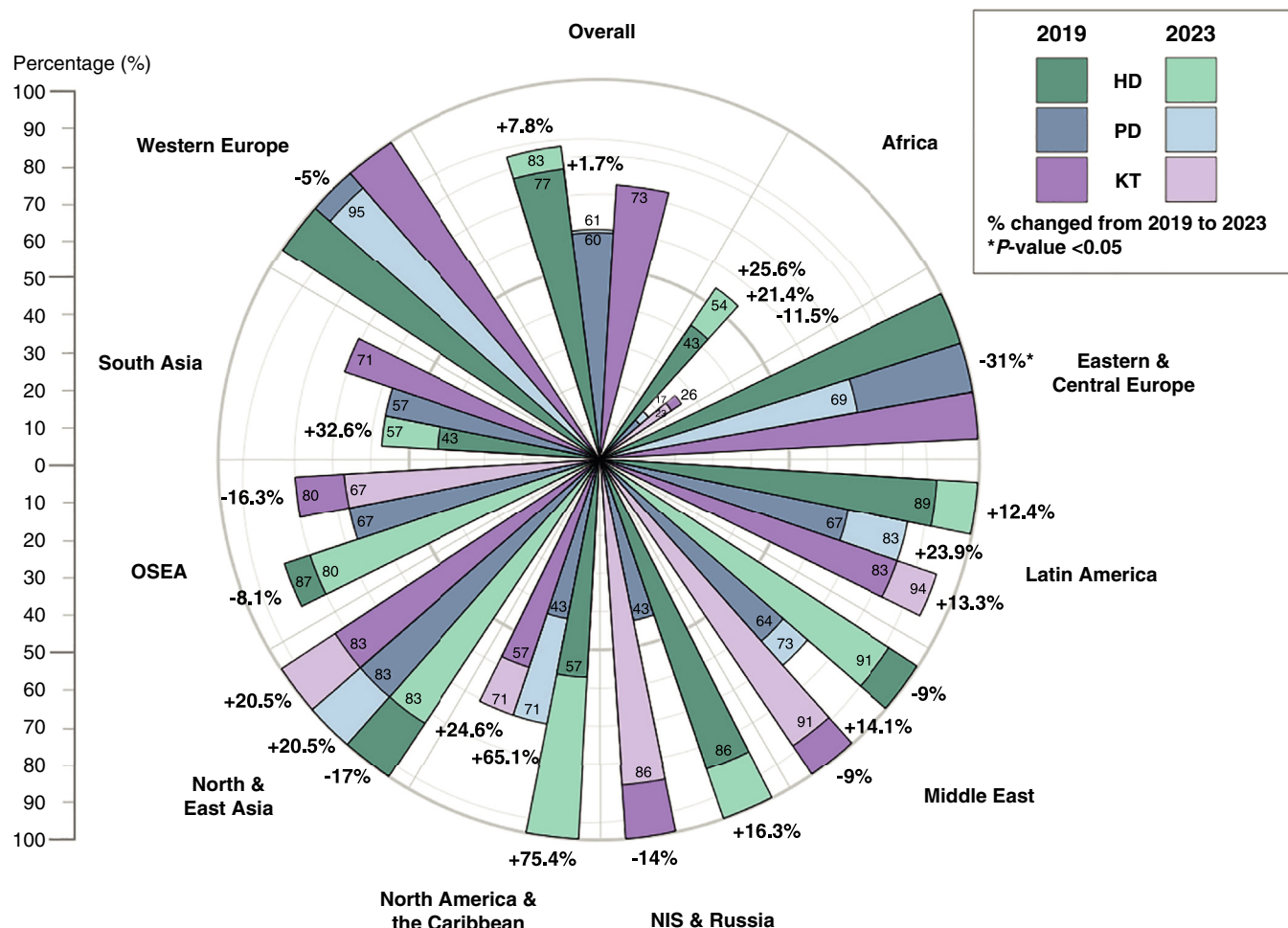


Fig. 73.18 Quality of kidney replacement therapy by International Society of Nephrology region changes noted pre- and post pandemic. Quality indicated by center-based hemodialysis service of adequate frequency (treatment three times a week for 3 or 4 hours) in hemodialysis (HD), peritoneal dialysis (PD) exchanges of adequate frequency (3–4 per day or equivalent cycles on automated PD) in PD, and appropriate immunosuppression and antirejection treatment in kidney transplantation (KT). *NIS*, Newly independent states; *OSEA*, Oceania and Southeast Asia.

CHANGES IN QUALITY OF KIDNEY REPLACEMENT THERAPY AND AVAILABILITY OF ROUTINE DIAGNOSTIC AND MANAGEMENT OF COMPLICATIONS ASSOCIATED WITH KIDNEY FAILURE

Overall, the proportion of countries where HD was offered thrice weekly (3 to 4 hours per session) increased by 7.8% (Fig. 73.18). This proportion increased in Africa (25.6%), Latin America (12.4%), Newly Independent States (NIS) and Russia (16.3%), North America and the Caribbean (75.4%), and South Asia (32.6%) (see Fig. 73.18). Overall, the proportion of countries where adequate PD (defined as three to four exchanges per day or equivalent cycles on automated PD) was offered increased by 1.7%. World regions with the highest percentage increase for adequate PD were North America and the Caribbean (65.1%), Latin America (23.9%), and Africa (21.4%). Overall, countries with capacity for provision of facilities for management of kidney transplant (defined as provision of appropriate immunosuppression and antirejection treatments) did not change and only increased in Latin America (13.3%), North America and the Caribbean (24.6%), and the North and East Asia region (20.5%) (see Fig. 73.18).

CHANGES IN ACCESS TO KIDNEY REPLACEMENT THERAPY

Globally, the change in the proportion of patients with access to KRT modalities, HD, PD, and KT were 2.8%, 50%, and –3.3% across both periods (Fig. 73.19). Countries where >50% of kidney failure patients can access HD remained the same in North America and Caribbean, OSEA, and the Western Europe region but increased in Africa (17.7%), Eastern and Central Europe (6.4%), the Middle East (22%), and North and East Asia (20.5%). Although there was a global reduction in the proportion of countries where >50% of kidney failure patients can access KT, there was an increase in Africa (100%), Latin America (29.4%), and Western Europe (41.7%). Only HICs reported an increase in proportion of countries with access to KT (1.8%); this proportion reduced in LMICs (–72.7%) and remained the same in LICs and UMICs (see Fig. 73.19).

CHANGES IN AVAILABILITY OF WORKFORCE FOR KIDNEY CARE

Overall, and among all countries that participated in both surveys, the prevalence of nephrologists increased by 30.4%

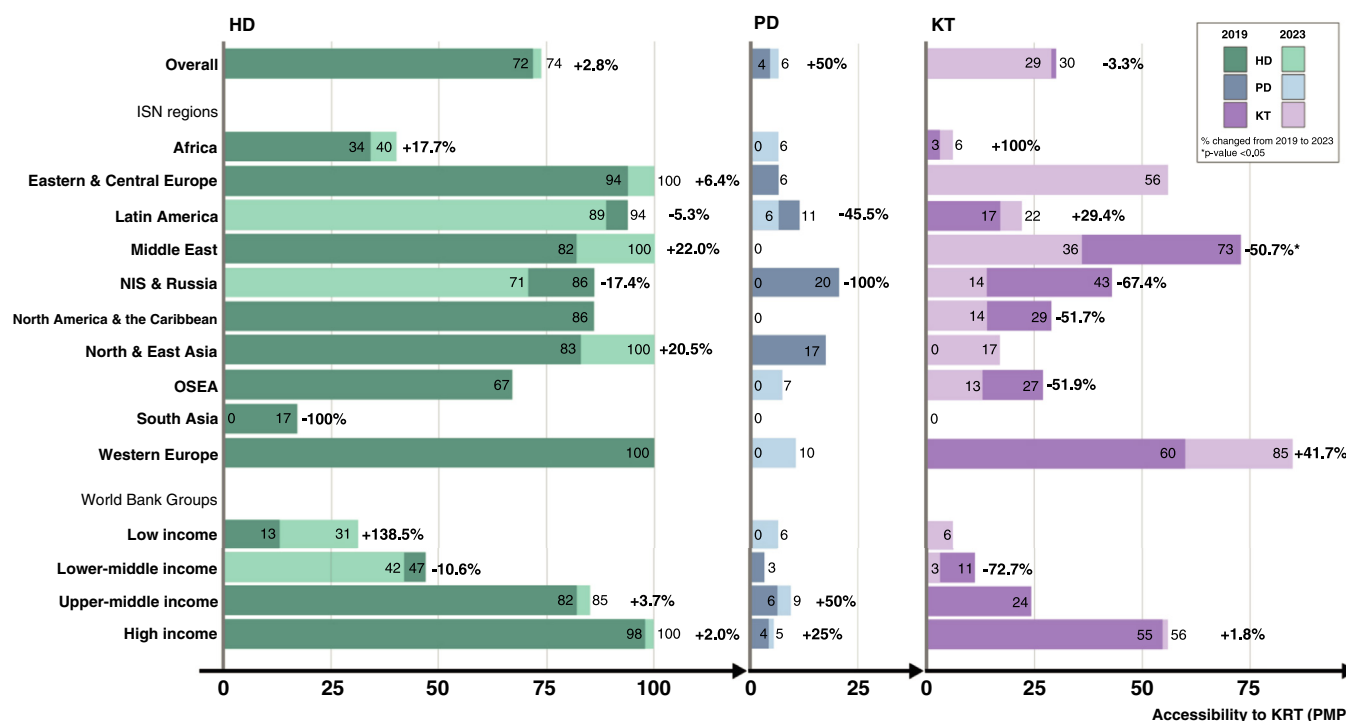


Fig. 73.19 Accessibility to kidney replacement therapy (KRT) by International Society of Nephrology (ISN) regions and World Bank Groups, pre and post pandemic. Accessibility was assessed as >50% of national population of kidney failure patients able to access hemodialysis (HD), able to start peritoneal dialysis (PD), and able to receive kidney transplantation (KT) in 2019 and 2023. Trends presented in percent change from 2019. *NIS*, Newly independent states; *OSEA*, Oceania and Southeast Asia; *PMP*, per million population.

(from 9.5 pmp in 2019 to 12.4 pmp in 2023). The Middle East (91.8%), Latin America (79.6%), and Africa (66.7%) had the highest proportion of increase in the density of nephrologists, and this was reduced in North America and Caribbean (−1%) and Eastern and Central Europe (−4.9%) (Fig. 73.20). When countries were stratified by economic ranking, nephrologist density increased across all income levels: LICs (15.4%), LMICs (38.5%), UMICs (35.1%), and HICs (10.6%). The proportion of countries that reported shortage of nephrologists decreased overall from 68 in 2019 to 62 in 2023 (−8.8%). Although Africa did not have a change in number of countries that reported shortages of nephrologists, it still had the highest proportion of countries (86%) with shortage of nephrologists as reported in the survey.

ISN-GKHA AND GLOBAL IMPACT IN KIDNEY CARE

The ISN-GKHA has demonstrated significant interregional and intraregional variability in the current capacity for kidney care across the world.^{51,80,81} Important gaps in the required infrastructure, services, and workforce for optimal kidney care delivery were identified in many countries and regions. The findings have implications for policy development toward establishment of robust kidney care programs, particularly for LLMIC. Basic infrastructure for services must be strengthened at the primary care level for early detection and management of AKI and CKD across all countries.^{7,32,86} Such information is crucial for advocacy with governmental and nongovernmental stakeholders to support countries

in improving the quality of kidney care. The ISN-GKHA provides a baseline understanding of where countries and regions stand with respect to several domains of the health system, thus allowing the monitoring of progress through the implementation of various strategies targeted toward improving kidney care. The ISN-GKHA reports are freely available online and will be updated every 4 years to develop country-specific “scorecards” and to facilitate identification of country-specific gaps and targets to enhance local kidney health.^{51,80,81}

GLOBAL INEQUITIES IN KIDNEY CARE

KIDNEY REPLACEMENT THERAPY AND THE CHALLENGES OF LIMITED ACCESS

Many barriers reduce access to dialysis around the globe.^{23,61,63,87-89} Unless dialysis is covered through state coverage or insurance schemes, access to dialysis is often delayed or limited by prohibitive out-of-pocket expenditures, patient age or gender, geography, and lack of expertise and infrastructure (Fig. 73.21).^{23,61,88}

KIDNEY REPLACEMENT THERAPY FOR END-STAGE KIDNEY FAILURE

The global median incidence of treated kidney failure reported in the ISN-GKHA was 146 (interquartile range, IQR, 107–213) per million population (pmp), but among countries reporting to the United States Renal Data System (USRDS) this varies significantly across the globe, with incidences ranging from 603 pmp in Mexico to 14 pmp in

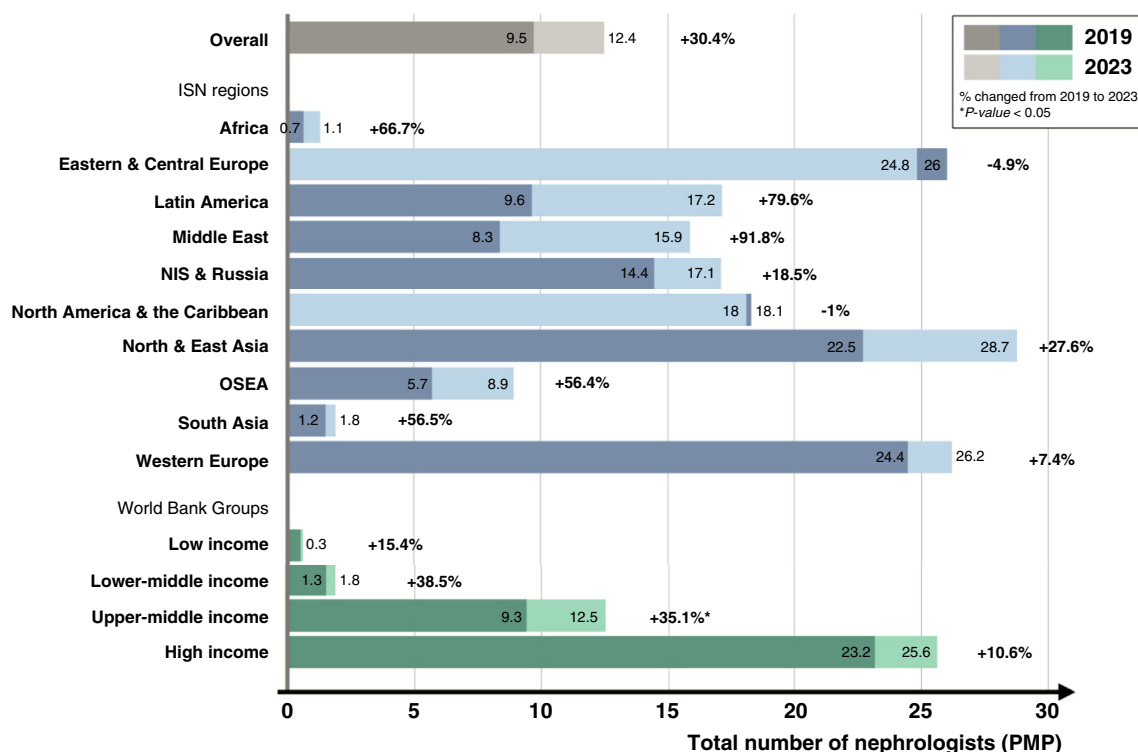


Fig. 73.20 Change in the distribution of nephrologists, pre (2019) and post (2023) pandemic era by International Society of Nephrology (ISN) regions and World Bank Income Groups. Trends presented in percent change from 2019. NIS, Newly independent states; OSEA, Oceania and Southeast Asia; PMP, per million population.

South Africa.^{51,90} The true incidence of ESKF in low- and lower-middle-income countries (LLMICs) is unknown as existing data come only from those who access dialysis and where data exists. Indeed, the incidence of ESKF in LLMIC may be higher than the prevalence, indicating high attrition rates predominantly because of inability to pay for dialysis or being too sick at presentation.^{90,91} The global median prevalence rate of ESKF reported in the GHKA was 823 pmp, although individual country prevalences range from 110 pmp in Bangladesh to 3839 pmp in Taiwan in the USRDS report, reflecting large global inequities in access to care.⁹⁰ In many LICs the prevalence is closer to zero. Further inequities exist when KRT is stratified by HD, PD, and KT, with many countries having no availability of PD or transplantation.⁵¹ Among children the global incidence and prevalence of kidney failure is unknown but is again highly variable depending on accessibility of chronic dialysis.^{13,61,92,93} Reported incidences in countries where KRT is available range from 4 to 13 per million age-related population (pmarp), and prevalences range from 22 to 77 pmarp.⁹⁴ Prevalence rates in children should ideally remain low as the goal everywhere should be to transplant children as soon as possible.

Large and persistent treatment gaps are evident in registries and the literature between those who need and those who obtain dialysis globally, with the largest gaps being in Africa (84%–91% not treated) and Asia (66%–83% not treated).⁶ It has been estimated that at least 2.1 million people died prematurely in 2010 because they could not access dialysis.⁶ These deaths occurred predominately in LLMICs. Other authors estimated a total of 1.2 million premature deaths resulted from lack

of access to KRT among people with hypertension and diabetes alone based on prevalence and expected rates of ESKF.⁹⁵ Of major concern is the fact that by 2030, attributable to population aging and increasing rates of diabetes and hypertension, the global need for dialysis is projected to double, with the greatest growth occurring in LLMICs.⁶

KIDNEY REPLACEMENT THERAPY FOR ACUTE KIDNEY FAILURE

Few data are available regarding the need for dialysis for AKI in LLMICs. In general, AKI appears more severe at diagnosis in LLMIC compared with HIC, most likely due to delays in presentation to hospital, and therefore the need for dialysis is higher, but access to dialysis is lower in LLMIC.^{7,23,25,96} Acute PD is an attractive modality for AKI because it is as effective as HD, requires far less infrastructure, and can be performed with solutions and catheters adapted to local resources.⁹⁷ PD should be cheaper than HD, especially if solutions are manufactured locally.⁹⁷ However, this may not be true because of the necessity to import fluids, customs, and corruption. For example, in Nigeria, the costs of 5 days of PD or HD for a child were similar and amounted to several times the monthly minimum wage in the country, making out-of-pocket payment impossible for most.^{98,99} In recent years, some governments have extended coverage for short-term dialysis for AKI (e.g., Ethiopia and South Africa) paid for by national health insurance schemes, inpatient government-funded services, or nongovernmental organizations and charities.^{100,101} If renal recovery does not occur within the coverage period for dialysis for AKI, however, costs revert to out-of-pocket payment, which is

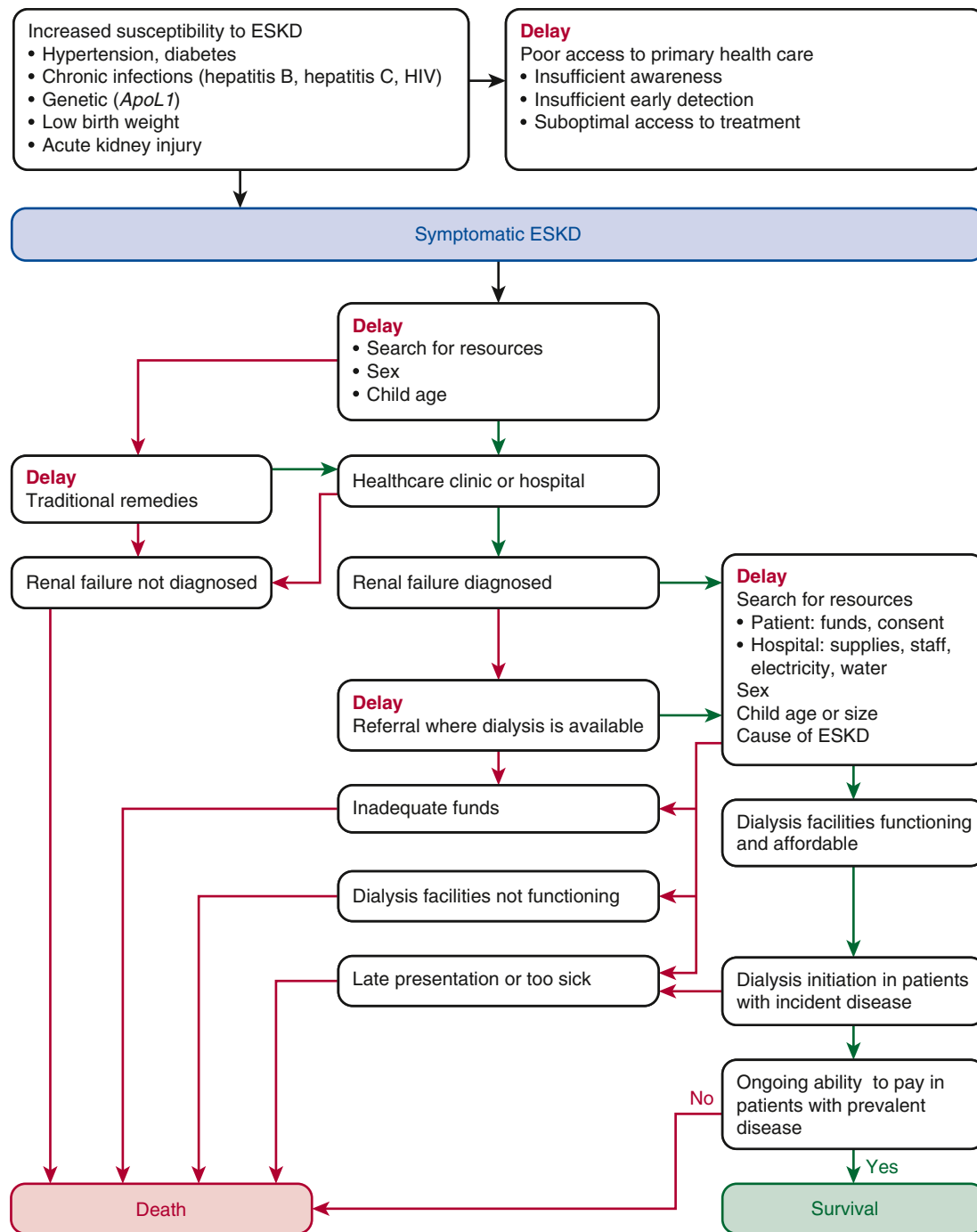


Fig. 73.21 Barriers to care in end-stage kidney disease in sub-Saharan Africa. Flow diagram showing barriers in access to dialysis contributing to mortality in end-stage kidney failure (ESKF) in many sub-Saharan African countries. Green arrows show factors present to facilitate diagnosis of ESKF or referral for or access to dialysis; red arrows show absence of these factors. Most barriers are related to access to care, access to diagnosis, out-of-pocket payments needed, and infrastructural resources.⁶¹ (From Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2016;4:e242–e250.)

unsustainable, and dialysis must be actively discontinued or the patient disappears.^{23,100}

There has been a global push for universal access to dialysis for AKI given that dialysis should be temporary because renal function can recover. It is therefore less expensive per patient than life-long treatment for ESKF.⁷ The Saving Young Lives [SYL] initiative, launched as a

combined endeavor among the International Pediatric Nephrology Association, ISN, International Society of Peritoneal Dialysis, and European Peritoneal Dialysis in 2012, has saved the lives of several hundred individuals (mostly children) and has trained and mentored local physicians and nurses to perform acute PD using home-made dialysis solutions and innovative use of alternative

Table 73.2 Suggestions for Preparation of Locally Mixed Peritoneal Dialysis Fluids^a**Dialysis fluid can be made by adding 50% dextrose to Ringers lactate, Plasmalyte B, or Hartmann solution**

- 1 L Ringers lactate + 30 mL 50% dextrose will make a 1.5% solution
- 1 L Ringers lactate + 90 mL 50% dextrose will make a 4.5% solution
- 1 L Plasmalyte B + 30 mL 50% dextrose makes a 1.5% dialysis solution
- 1 L Plasmalyte B + 90 mL 90% dextrose makes a 4.5% dialysis solution
- 1 L Hartmann + 30 mL 50% dextrose makes a 1.5% dialysis solution
- 1 L Hartmann + 90 mL 50% dextrose makes a 4.5% dialysis solution

Using 0.45% saline + 5% dextrose + NaHCO₃

- 1 L Normal saline + 40 mL 50% dextrose + 40 mL 8.4% NaHCO₃ + 60 mL 3% NaCl will make a 1.5% dialysis solution
- 1 L Normal saline + 60 mL 50% dextrose + 40 mL 8.4% NaHCO + 60 mL 3% NaCl will make a 2.5% dialysis solution

Using 0.9% saline + 5% dextrose + NaHCO₃

- 1 L Normal saline 1 L 5% dextrose 100 mL 8.4% NaHCO₃ makes a 2.5% dialysis solution

Please note the final electrolyte and glucose concentrations of the above solutions:

- 1 L Plasmalyte B/Ringers lactate/ Hartmanns + 30 mL 50% dextrose (15 g) will generate a solution with the following concentrations:
glucose = 1.45%, Na = 126 mmol/L, HCO₃⁻/Lactate = 27 mmol/L, K = 3.8 mmol/L, Mg = 1.45 mmol/L, Osmolality = 342
- 1 L Normal saline + 40 mL 8.5% NaHCO₃ (40 mmol) + 40 mL 50% dextrose (20 g) + 60 mL 3% NaCl (30 mmol) will generate a solution with approximately the following concentrations: Na = 130 mmol/L, HCO₃ 35 mmol/L, glucose = 1.7%, osm = 345
- 1 L Normal saline 1 L 5% dextrose + 100 mL 8.4% NaHCO₃ makes a 2.5% dialysis solution will generate a solution with the following concentration: glucose = 2.38%, Na = 121 mmol/L, HCO₃ = 48 mmol/L, osm = 374

^aWhen adding more glucose, the electrolyte concentrations will change slightly but not significantly.
Reproduced from 2021 International Society of Peritoneal Dialysis (ISPD) Guidelines/recommendations for PD for AKI.¹⁰³

catheters (Table 73.2).¹⁰²⁻¹⁰⁴ This program, however, remains dependent on external funding and is highly reliant on individual champions. Unexpected challenges outlined by those implementing the SYL program include lack of access to routine laboratory testing, staff changes, unreliability of financing sources, cultural differences, and lack of referrals to SYL centers.¹⁰⁵ These challenges highlight the crucial importance of program integration into the existing health system structures, which is important to avoid generation of vertical programs and contribute to strengthening the health system.

KIDNEY TRANSPLANTATION

Transplantation is by far the most cost-effective KRT solution in the long term, but if costs of the surgery and long-term medication and follow-up are not sustainably covered by governments, health insurance, or donors, accessibility and success are significantly reduced.^{63,106-108} In India, for example, 72% of patients undergoing kidney transplantation reported some form of resulting financial crisis, with 10% expressing regret at having embarked on the procedure.¹⁰⁹ Globally, the majority of transplants are conducted in HICs in part due to lack of resources and knowledge in lower-income countries, as well as cultural practices and absence of legal frameworks governing organ donation (see Fig. 73.19).^{51,110} Worldwide, transplant activity is directly associated with health expenditure, physicians per capita, and life expectancy at birth and is inversely associated with reliance on out-of-pocket payments and reliance on external resources for health.¹¹⁰ Transplantation can be successful in lower-resource settings, but this requires major commitment to sustainable funding and quality of care.⁶³ In some countries where the local population cannot access KT, where resources exist, patients and donors may travel abroad for transplantation.^{111,112} Organ trafficking and commercial transplantation are a global problem because of the ethical dilemmas they pose and their association with worse outcomes for the transplant recipient. They put living donors at risk in low-income environments, where they are unlikely to have been well screened and cannot access routine follow-up.¹¹³⁻¹¹⁵ How to approach kidney transplant recipients who return from another country with suspicion of having bought a kidney is an ongoing ethical challenge.^{116,117}

Clinical Relevance: Catastrophic Health Expenditure

Out-of-pocket health expenditure amounting to >10% of total household expenditure or 40% of household nonfood expenditure.¹⁰⁹ Such expenses risk plunging families into poverty and reduce their capacity to meet other necessities.

Clinical Relevance: Universal Health Coverage

Defined by the World Health Organization as “all people receiving quality health services that meet their needs without being exposed to financial hardship in paying for the services.”¹⁷¹ The three dimensions of UHC include expanding priority services, expanding the number of people covered, and reducing out-of-pocket payments and thereby financial hardship.

Clinical Relevance: Opportunity Costs

Health gains that could have been realized should the money have been allocated to an alternative intervention.

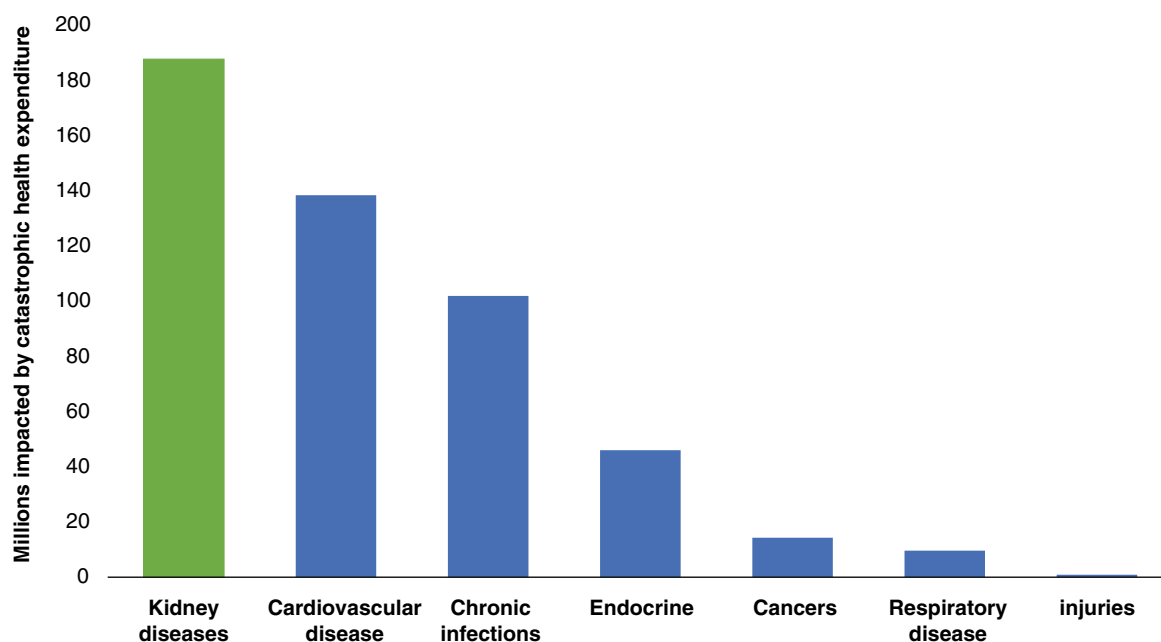


Fig. 73.22 Leading global causes of catastrophic health expenditure in low- and middle-income countries. Millions of people impacted by catastrophic health expenditure per health condition. Catastrophic health expenditure—total out-of-pocket health expenditure measured as an (impoverishing) percentage of household resources. (Data derived from reported rates obtained in systematic review multiplied by population at risk <http://dcp-3.org/disease-control-priorities>, Chapter 6; Essue BM, Laba M, Knaul F, et al. Economic burden of chronic ill health and injuries for households in low- and middle-income countries. In Jamison DT, et al., eds. *Disease Control Priorities: Improving Health and Reducing Poverty*, 2017, The International Bank for Reconstruction and Development/The World Bank © 2018 International Bank for Reconstruction and Development. Washington, DC, USA: The World Bank; 2018.).

Dilemmas posed by lack of sustainable kidney replacement therapy in low-resource settings

Attempting to meet dialysis costs out of pocket generally constitutes a catastrophic health expenditure for a patient and family.^{88,118} Kidney diseases including dialysis and costs for basic medications are the leading cause of catastrophic health expenditure across LLMICs (Fig. 73.22).¹¹⁹ Protection from individual financial risk is one of the cornerstones of the global push for UHC, but how far a government should go to meet the needs of patients requiring dialysis potentially at the expense of funding other programs is a contentious issue.^{120,121} The limited health budgets and current burdens of renal and nonrenal disease in LLMIC make progress toward universal access to KRT challenging.¹²¹ Globally, on average, 0.1% of the population requires treatment for ESKF at a cost of around 2% to 4% of health expenditure.³⁵ These direct costs of ESKF often do not account for costs of frequent hospitalizations and management of other comorbidities, so total costs are even higher.^{35,122} Such disproportionate expenditure on a relatively small proportion of the population poses a challenge in terms of prioritization of health expenditure and opportunity costs when budgets are limited. Some governments (e.g., Guinea, Cameroon, Senegal, and India) alone or in partnership with private institutions or charities have elected to subsidize the cost of chronic dialysis, requiring small out-of-pocket payments.^{118,123-125} Even these prices, which are much lower than the average unsubsidized cost of USD 60 to 100 for a single HD treatment, remain unsustainable for families long-term and often do not cover the vital costs of transport, medications, or time lost from work which can almost double the total out-of-pocket cost for a patient.^{88,118,126,127} In LMICs a large proportion of patients who initiate chronic dialysis cannot sustain it and stop dialysis within days to weeks.^{61,127,128}

Even among European cohorts, the mortality of children on dialysis was related to public health expenditure.¹²⁹ Sustainable and sufficient funding is therefore necessary for successful treatment of kidney failure.¹²¹ It is unrealistic to imagine that within the short to medium term, all people requiring KRT will be able to access it equitably. Recognition of this fact is important to ensure that patients are still treated with dignity and that they and their families have universal access to symptom relief and palliative care¹³⁰ (see Chapter 61).

MORAL DISTRESS AMONG NEPHROLOGISTS WHERE ACCESS TO KIDNEY CARE IS LIMITED

Moral distress is a specific psychologic response to morally challenging situations such as those of moral constraint or moral conflict or both.¹³¹ Factors associated with moral distress in clinicians include a poor ethical climate, lack of interdisciplinary collaboration and peer support, work load (high and low patient loads), working in acute care, feeling of disempowerment, low levels of health care worker autonomy, and lack of opportunities for ethical debate.¹³² Moral distress is associated with reduced job satisfaction and engagement, higher risk of burnout, poorer patient outcomes, and reduced patient satisfaction.^{133,134} In resource-limited settings, moral distress arises from too little care being available for people who would otherwise benefit considerably, rather than grappling with demands for provision of “futile” care, which may occur in HICs (e.g., in the severely ill or frail). The amount of care delivered in resource-limited settings is often directly proportional to the financial resources of the patient and/or the institution, and little progress has occurred in recent years. For example, the proportion of persons with kidney failure receiving hemodialysis in low-income regions when

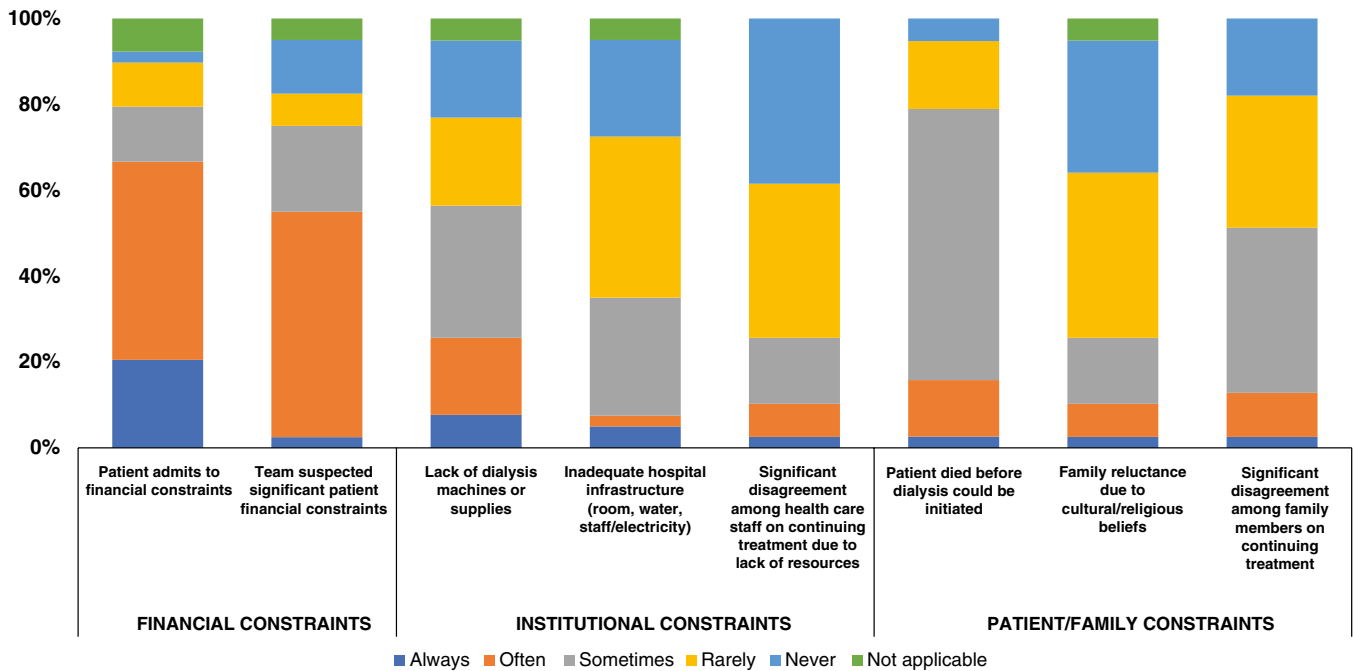


Fig. 73.23 Frequency of constraints limiting access to kidney care in sub-Saharan Africa. Frequency with which constraints to accessing kidney care were encountered within the previous 2 years by 40 senior nephrologists in sub-Saharan Africa. (Data compiled from Ashuntantang G, Miljeteig I, Luyckx VA. Bedside rationing and moral distress in nephrologists in sub-Saharan Africa. *BMC Nephrol.* 2022;23[1]:196.)

paid for solely out of pocket has increased from 13% to 19%, while it has remained stable at 6% for lower- to middle-income regions.⁵¹ Most people who require dialysis in these settings are therefore unable to access it, which is a source of moral distress for the clinicians but also for families who must decide to decline unaffordable care.¹³⁵

In regions where access to care is limited, moral distress is common. In sub-Saharan Africa, among 40 nephrologists surveyed, the majority saw patients weekly in need of dialysis whom they could not dialyze.¹³⁵ When dialysis was provided, clinical compromises were common (e.g. reducing session frequency or using temporary catheters). Most (66%) of the nephrologists reported frequent lack of basic diagnostics and medication, while >80% reported high out-of-pocket expenses for patients. Several patient, disease, and institutional factors influenced who gained access to dialysis (Fig. 73.23). Patient financial constraints and poor chances of survival limited the likelihood of receiving dialysis, while a patient being the family bread winner increased the likelihood. Patient and institutional constraints resulted in most nephrologists frequently having to make difficult choices, sometimes having to choose between patients in terms of who received life-saving dialysis and who did not. Most nephrologists always, often, or sometimes felt burdened by ethical dilemmas and worried about patients during out-of-office hours, which are signs of moral distress.¹³⁵ Almost 46% of nephrologists reported frequently regretting their choice of profession and 26% considered leaving the country. Being forced to provide second best or no treatment for conditions such as kidney failure, which could easily and cheaply have been prevented through public health measures and access to quality and affordable primary care, is stressful and can be a driving force for highly trained clinicians (doctors and nurses) wanting to leave the profession and even the country, contributing to “brain drain.”¹³⁵

POLICY DEVELOPMENT AND ACCESS TO KIDNEY REPLACEMENT THERAPY

Most countries in LLMIC cannot afford to provide universal access to KRT. However, if states elect to provide some KRT, whether through sole funding or through partnership agreements, it is important that transparent and just eligibility criteria governing who will gain access to KRT are established.^{120,136} Existence of such criteria should facilitate bedside discussions and decision making, which should involve consideration of both clinical and social/environmental realities that may impact a patient’s ability to continue dialysis without incurring catastrophic health expenditures. Official criteria will also reduce moral distress among health care workers, taking full responsibility for individual case-by-case life and death decision making from individual clinicians’ shoulders.¹³⁵ Eligibility criteria may be more flexible for AKI than ESKF as the personal/institutional resources required will be less for AKI. To be effective, however, the development of such guidelines requires a transparent, consultative, and iterative development process, as well as engagement with many different stakeholder groups. Potential policy scenarios and processes for development of dialysis guidelines are outlined elsewhere.^{120,137} Policy options would include 1. provision KRT to all, 2. provision of dialysis for AKI only, 3. provision of KRT to a restricted eligible group (i.e., rationing), and 4. no state provision of KRT and allowing market forces to determine access (Table 73.3). Major inequities are inherent to each of these scenarios: Where universal KRT is not covered, inequities exist between countries; where AKI only is covered, inequities exist between those with AKI and those with ESKF; where rationing occurs, inequities exist between those meeting and not meeting the eligibility criteria; where

Table 73.3 Examples of State Approaches to Provision of Kidney Replacement Therapy in Lower-Resource Settings

Expected Benefits	Potential Harms	Impact on Equity	Expected Cost	Practical Issues	Cautions	Burdens for Patients
Universal coverage of dialysis for AKI and ESKD	Potential for all to benefit	Diversion of funds to kidney disease, away from other diseases	Good for patients with kidney failure	Expensive	Reduce dialysis frequency to reduce costs and increase capacity	Patients may need to move for dialysis, lose jobs, etc. Transport (and medication) costs not covered
Government subsidy for dialysis	Potential for all to benefit	Kidney disease prevention programs often not in place	Expensive	Infrastructure limited; in effect, dialysis may not be available to all	Donations may drive program	Large out-of-pocket expenses remain (\$10-\$20 per treatment, medication, transport)
Universal coverage for AKI only	Saves lives of patients with AKI	Tends to favor those with resources to pay	Expensive	Periodic stock outs when companies decline to deliver supplies because of late payments	Dialysis twice a week	Out-of-pocket expenses
State coverage under limited conditions	Benefit for those eligible to receive treatment	ESKF patients not dialyzed	More cost-effective than dialysis for ESKF, short-term treatment, may not require much infrastructure	Dialysis not available everywhere	If AKI does not recover, must withdraw dialysis	Anxiety, distress, fear, helplessness, lack of understanding
No state coverage	Limited to rich few	May exacerbate inequities, especially for poor, vulnerable, sick	Prioritize peritoneal dialysis (cheaper)? Prioritize transplant (most cost-effective)?	Patient selection process may impose delays Need good palliative care services	Need clear transparent guidelines, community engagement, appeal possibilities	If not fully informed, poor patients may not realize life-long expenditure required

Adapted from^{100,121}.

Table 73.4 Health System Components Important for Kidney Care

Building Block	Opportunity to Improve Kidney Health
Leadership and governance	• Resource allocation: • Prevention versus treatment • Primary secondary tertiary level care • Dialysis, transplantation yes/no • Priority setting—strategies for NCDs, kidney disease • Public health (primordial prevention) • Intersection between all sustainable development goals and health system • Legalize transplantation, brain death • Legalize cadres of health care workers, scope of practice • Models of care • Transparent guidance to allocate scarce resources • Reduce monopolies • Tackle corruption
Health financing	• Sustainability • Universal health coverage • Domestic resource mobilization • Transparent procurement strategies and negotiations • Fair remuneration for health care workers
Health information systems	• Track risk factors for kidney disease • Disease registries—AKI, CKD, dialysis, transplant living donors • Track costs—health system and societal • Track quality of care
Health workforce	• Appropriate training for all cadres • Reduce brain drain • Reduce moral distress • Incorporate, engage traditional health care workers
Medical products, vaccines, and technologies	• Essential medicines • Essential diagnostics • Local manufacture of health products • Extend essential diagnostics and medicine lists • Access to products and technologies for dialysis and transplantation
Service delivery	• Optimize infrastructure • Optimize access to early diagnosis • Optimize access to sustainable, affordable, accessible early treatment • Optimize access to dialysis and transplantation • Ensure quality care • Safe, effective, timely, patient-centered, efficient, equitable • Resilient

AKI, Acute kidney injury; CKD, chronic kidney disease; NCD, noncommunicable disease.

Reproduced from Ulasi II, Awobusuyi O, Nayak S, et al. Chronic kidney disease burden in low-resource settings: regional perspectives. *Semin Nephrol.* 2023;42(5):151336.

dialysis is not covered by the state or only partially subsidized, inequities exist within countries and regions depending on ability to pay and location. In addition, when developing KRT policies and guidelines, consideration must also be given to realistically achievable minimum safe standards of practice, as patients and providers may attempt to reduce dialysis frequency and dose to curb costs at the expense of health.¹³⁸ To improve financial sustainability and quality of care, public-private partnerships may be a feasible strategy to expand access and maintain quality of care.¹³⁹

It is also important to recognize, however, that other inequities and health challenges in many LMICs may be more urgent and affect many more people than kidney disease.^{120,121} Individual financial risk protection is, however, one of the three dimensions of UHC; therefore dialysis provision poses a dilemma—people die without it, people are impoverished because of it, and opportunity costs are extremely high.¹²¹

Priority setting within each country is therefore essential but requires reliable relative assessments of the disease burden, which is lacking for kidney disease in most LMIC. It is crucially important to advocate for the need for prevention, early detection, and treatment of AKI and CKD because these should be achievable in primary care settings at much lower cost than KRT, and over the long term it will be the most effective means to reduce health expenditures and the need for KRT.^{37,121}

PUBLIC HEALTH INITIATIVES TO IMPROVE KIDNEY HEALTH AND ACCESS TO CARE

The public health approach to disease control must consider action at all levels of the health system and beyond (Table 73.4).¹⁴⁰ At the macrolevel, government health policy, and intersectoral collaboration are key to improving

Table 73.5 Levels of Public Health Interventions, Specific Steps, and Potential Impact on Kidney Health

Elements	Specific Steps	Impact on Kidney Health	Comments
Socioeconomic factors	Improved livelihood, education, and sanitation Universal access to timely affordable health care	Overall improvement in health Reduced NCD burden	Requires political will linked to overall development and economic growth over long term Radical redesign of health systems required Requires political will likely
Changing the context to make default decisions healthy	Reducing salt in packaged foods Higher taxation for tobacco, sweetened beverages, and processed foods Regulate OTC/indigenous medicine use Mandate protective clothing/footwear in certain occupations (outdoor exposure) Access to quality reproductive care, legalization of abortions Widespread ORS availability Vaccinations, improved immunization coverage for VPDs	Reduction in kidney disease risk Factors (hypertension, diabetes, metabolic syndrome) Reduction in CKDnT Slowed progression of CKD Reduction in occupation-related CKD and AKI Reduction in obstetric AKI	Opposition from special interest groups and industry lobby
Long-lasting protective interventions	Kidney disease screening at schools and workplaces Vector control	Reduced infection-related kidney disease (AKI and CKD) Reduced infections in those with kidney disease Early-prevention opportunities	Can be controversial Requires multisectoral approaches
Clinical interventions	Treatment of risk factors and predisposing conditions Measures to improve adherence to treatment	Reduction in CKD development and progression Improvement in obstetric AKI through tracking outcomes	Needs manpower and resources Ensuring a continuum of care Effective health information systems deployment Least effective
Counseling and education	Increase kidney disease awareness Dietary counseling Increase physical activity	Can slow CKD progression and prevent AKI in certain situations	Requires civil society engagement and ownership

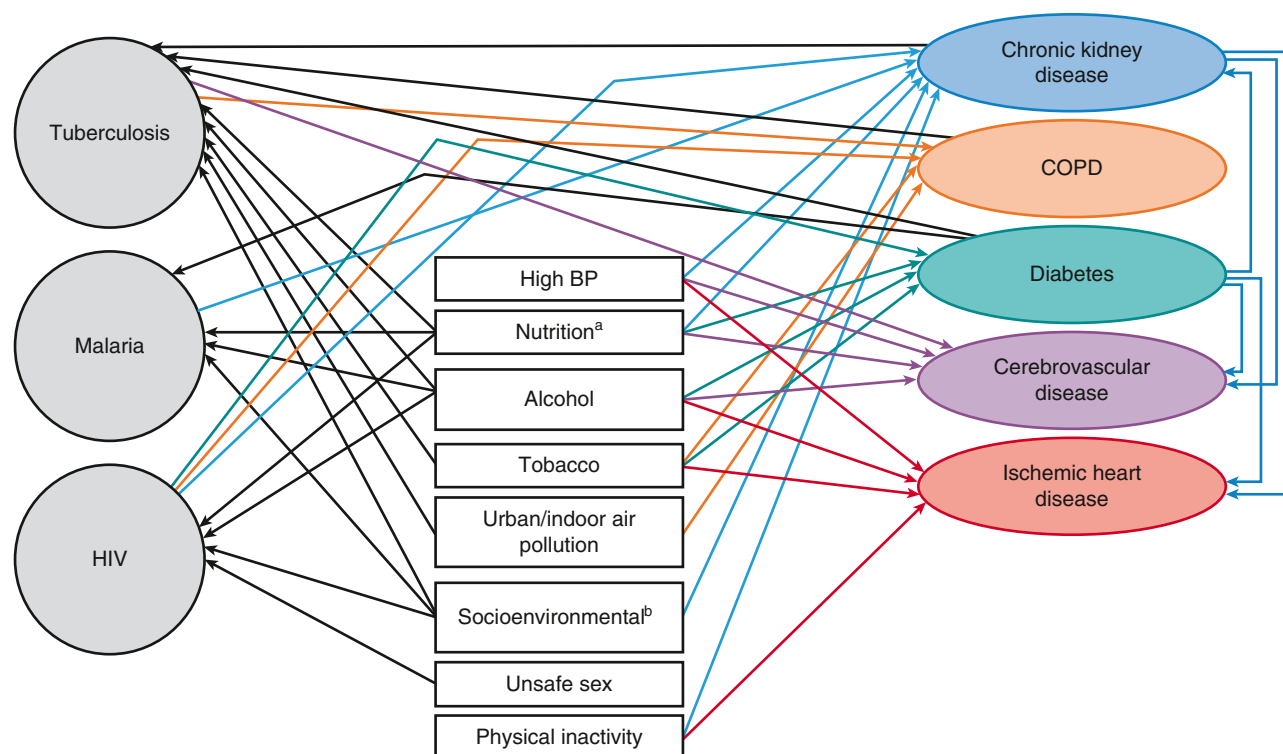
CKDnT, Chronic kidney disease of nontraditional causes; ORS, oral rehydration solution; OTC, over the counter; VPD, vaccine-preventable disease.

Reproduced from Joshi R, John O, Jha V. The potential impact of public health interventions in preventing kidney disease. *Semin Nephrol.* 2017;37(3):234–244.

population health; at the mesolevel, health administrators must translate policies into action and conduct monitoring and evaluation of exercises to determine effectiveness and address problems; at the microlevel, individual health care workers and patients must work together to preserve health. Kidney disease is impacted by many social and structural factors and therefore is amenable to public health interventions. Government policies that have proven effective in reducing the risk of NCDs include economic incentives to reduce prices of healthy food, taxation on unhealthy foods and beverages, school programs including health and physical education, food advertising restrictions and standards, increasing safe spaces for sport and recreation, media campaigns promoting smoking cessation, health warnings on cigarette

packages, restriction of tobacco advertising, increased taxes on tobacco, and prohibition of smoking in public spaces.^{29,141,142} Implementation of public health programs such as vaccinations, health education, malaria prevention, providing access to clean water, antenatal care, ensuring safe deliveries, and optimizing child nutrition can all also reduce the burden of AKI and CKD in the community (Table 73.5).^{14,29,141,143} The risk of kidney disease begins in fetal life and spans the life course, so prevention of kidney disease starts with optimization of health and education for girls, UHC for antenatal care and safe deliveries, and optimizing child nutrition and growth.⁶⁷

Targeted screening of high-risk individuals (with diabetes, hypertension, and CVD) has been shown to be cost-effective in detecting and treating CKD early and



^aEncompasses underweight, overweight/obesity, low fruit/vegetable consumption, high glucose intake

^bConditions associated with informality: overcrowding, unsafe water, & sanitation

Fig. 73.24 Interaction among tuberculosis, malaria, and human immunodeficiency virus and risk factors /disease precursors and noncommunicable diseases. BP, Blood pressure; COPD, chronic obstructive pulmonary disease. (Adapted with permission from Oni T, Unwin N. Why the communicable/non-communicable disease dichotomy is problematic for public health control strategies: implications of multimorbidity for health systems in an era of health transition. *Int Health*. 2015;7[6]:390–399.)

delaying progression to ESKF.^{33,35,144,145} Screening and early treatment strategies require availability of the necessary resources and trained personnel at all levels of the health system to manage kidney disease.^{33,146} Given the broad reach required and the need for simple screening strategies, which can be integrated into existing WHO NCD strategies, task-shifting may be required and desired to improve uptake and coverage, especially in countries with large populations such as India.^{147,148} Systematic screening activities will also facilitate generation of renal registries and provide information on local kidney disease burdens necessary for priority setting and program planning.^{94,149} Integration of care for patients with kidney disease within communicable and noncommunicable disease programs, coupled with community engagement and empowerment of patients, promises to improve quality of care with greater efficiency and at lower cost, as well as provide a more holistic approach to patient care (Fig. 73.24).^{39,150} Attention must be paid to quality care, which requires all aspects of care delivery to be optimized: access, acceptability, affordability, adequate resourcing, education of health care personnel, appropriate guidelines being in place, monitoring and evaluation of services and outcomes, and empowerment and education of patients themselves.^{137,151–154}

HEALTH IN ALL POLICIES AND THE SUSTAINABLE DEVELOPMENT GOALS

The implementation of population-level strategies to reduce the risk of NCDs and kidney disease requires coordinated multisectoral action across government and society, as emphasized in the Health in All Policies strategy.^{155–157} The Health in All Policies framework reflects the principles of “legitimacy, accountability, transparency, and access to information, participations, sustainability and collaboration across sectors and levels of government” as important components of an equitable approach to improving population health.¹⁵⁶ This multisectoral approach highlights the impact of the nonhealth sector (e.g., industry, education, housing, transportation, agriculture, and environment) in improving population health. In recognition of this multisectoral impact on health and the negative impact of illness on communities and overall economic growth, the United Nations launched the Sustainable Development Goals (SDGs) in 2015 with the goal of achieving healthy people living on a healthy planet.^{158,159} Multiple social and structural factors impact the risk of kidney disease across the life course including poverty, overnutrition or undernutrition, poor education,

Table 73.6 Climate and Disaster-Associated Kidney Health Risks

Climate Factor	Kidney Impact	Reference
Heat	↑ Stones	17,173
	↑ AKI	
	↑ Urinary tract infections	
	↑ CKD mortality	
	↑ Low birth weight, preterm birth	16,174
	↑ Preeclampsia	
Air pollution (PM _{2.5})	↑ Gestational diabetes	
	↑ CKD	175
	↑ End-stage kidney failure	
	↑ Transplant rejection	176
	↑ Transplant loss	
	↑ Global mortality	
Drought	↑ Low birth weight, preterm birth	16,174
	↑ Preeclampsia	
	↑ Food scarcity, malnutrition	177,178
	↑ Displacement of peoples	
Floods	↑ treatment interruptions	179
	↑ Infections, diarrhea	180
	↑ low birth weight, preterm birth	16
	↑ Preeclampsia	
Infections	↑ Treatment interruptions	179
	↑ Malaria, Dengue	178
Earthquakes	↑ AKI	179
	↑ Treatment interruptions	

AKI, Acute kidney injury; CKD, chronic kidney disease; HR, adjusted hazard ratio; RR, relative risk.
Adapted from Luyckx VA, Klimawandel LN. *NEPHRO-News*.

access public health interventions and health care, gender inequality, environmental and occupational factors, urban or rural location, violence, social disruptions and war, climate change, and the effects of globalization. The impact of climate change on kidney health is outlined in Table 73.6. Each SDG has direct or indirect relevance to kidney health (Fig. 73.25); therefore progress toward each SDG promises to improve the global burden of kidney disease.^{36,141} It is important to acknowledge the linkages with other diseases. Therefore the nephrology community has the opportunity to lead in working with public health professionals and all levels of health care workers on the ground to improve synergies and coordination between disease silos to improve overall health and thereby kidney health.

PRIORITY SETTING AND HEALTH TECHNOLOGY ASSESSMENT

As discussed earlier, reliance on out-of-pocket payment for KRT and rationing of access to dialysis are highly inequitable and contribute to many annual deaths globally.^{6,136,160} The costs of dialysis and transplantation are high for all health

systems, and even in a HIC such as the United States, which otherwise does not have UHC, the need to create an exception and provide coverage for ESKF patients has long been recognized and implemented.¹⁶¹ The challenge, however, is how to provide equitable access to KRT, sustainably and without compromising funding for other health priorities. Systematic analysis of local circumstances in terms of disease burden, projected costs (direct, indirect and opportunity costs), and expected benefits is required before developing policies around provision of KRT. Such analyses are all too often not performed, leading to the provision of inadequate services that may exacerbate inequities.¹⁰⁰ Priority setting is the process of resource allocation when budgets are limited and ultimately results in decisions to include or exclude specific health conditions or interventions.³⁷ Cost-effectiveness cannot be the only criteria, as people with expensive-to-treat conditions such as kidney disease tend to be deprioritized.¹⁰⁰ Equity must be a consideration.¹³⁷ Transparent priority setting requires reliable data on disease burden, availability of effective and acceptable therapeutic strategies, knowledge of the costs of coverage of these interventions, and a plan to progressively reach equitable distribution and access.^{120,121} Engagement with health care workers and communities in the priority-setting process is important for transparency, to establish accountability, and to ensure understanding, especially if access will not be possible for all initially.^{120,136} Many inequities were laid bare during the pandemic, and a global reckoning is required going forward to ensure more equity and more accessibility of health care for all including the spectrum of kidney care.^{37,162-164}

The example of Thailand's approach to provision of KRT under UHC is highly instructive of this priority setting process and ongoing community engagement. Thailand is a middle-income country and introduced UHC in 2002.¹⁶⁵ At that time, coverage for KRT was deliberately excluded because of the high anticipated costs, which led to significant inequity between people with additional health insurance and those without catastrophic health expenditures among those who tried to pay themselves.^{166,167} After systematic investigation and in the context of some favorable political circumstances, KRT was included under UHC in 2008.¹⁶⁸ Before implementation, much consideration was given to cost containment including a PD-first policy (analyses had shown that PD was more cost-effective than HD); bulk negotiation for pricing of dialysis supplies and medications (e.g., 25% reduction in cost of HD and 62% reduction in cost of erythropoietin) and negotiated incentive and reimbursement strategies.^{166,168} PD centers of excellence providing training and support to peripheral centers were established, and active stakeholder, community, and patient engagement was conducted to enhance uptake and provide support in communities to meet additional needs such as transport costs.¹⁶⁶ The number of patients requiring KRT initially grew rapidly as services expanded but has since plateaued, and outcomes are comparable with those of other programs.^{166,169} This program, successful for >15 years, is now facing a push for patient choice of dialysis modality, as well as potential lobbying for HD from various stakeholder groups, with the result that more patients are choosing HD.¹⁷⁰ This is anticipated to increase costs of the KRT program substantially and is highlighting the critical importance of prevention of kidney disease to contain costs.



Fig. 73.25 Sustainable Development Goals. Each one has relevance for kidney health. Icons reproduced according to guidelines as stipulated at <https://www.un.org/sustainabledevelopment/news/communications-material/>

CONCLUSION

The global burden of kidney disease has been relatively underappreciated until recent years. International activities are mounting to better quantify this burden, detect and address gaps in kidney care, and advocate for patients unable to access KRT. Collaboration is required to ensure that countries and regions facing greater challenges are supported to strengthen their health systems, implement public health strategies to prevent kidney disease, and progressively achieve equitable access to kidney care.

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